



Escuela Colombiana de Ingeniería Julio Garavito

Redes de computacion 2026-1

Laboratorio VI

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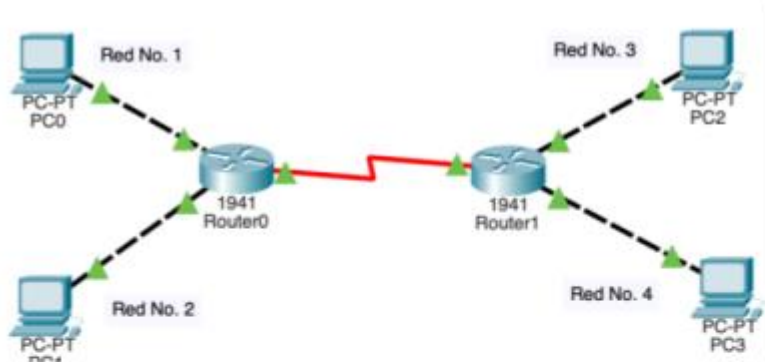
Simulations

Conduct the following tests individually or in groups, as instructed by the professor, and document your experience

1. Router Simulation

[For groups of 1, 2, and 3 students]

Open Packet Tracer and create a network following the diagram.

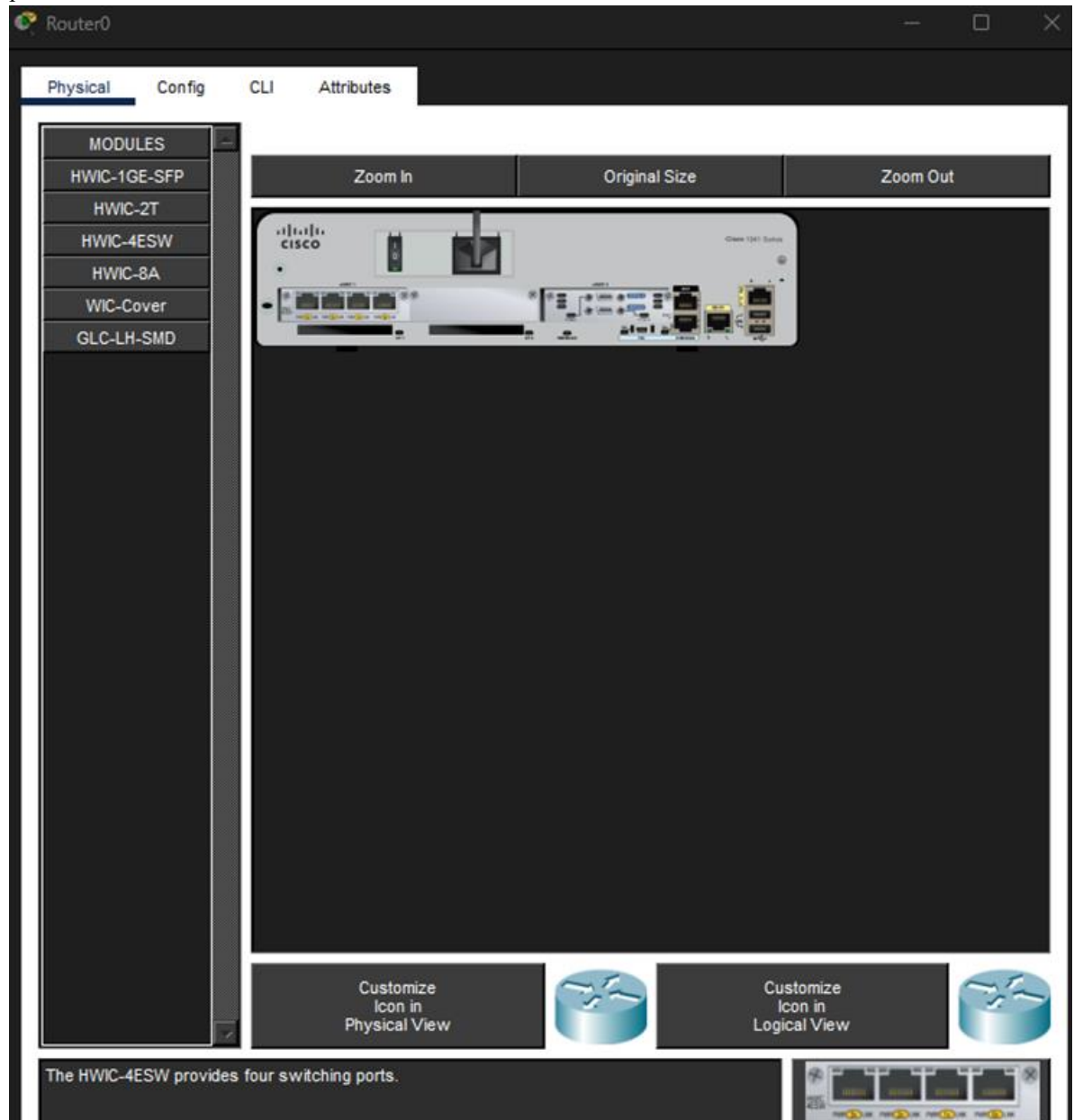


Now, to configure it, connect from a console similarly to how you do with switches. (The blue cable is a console cable and connects to the serial port – RS232 of each device).

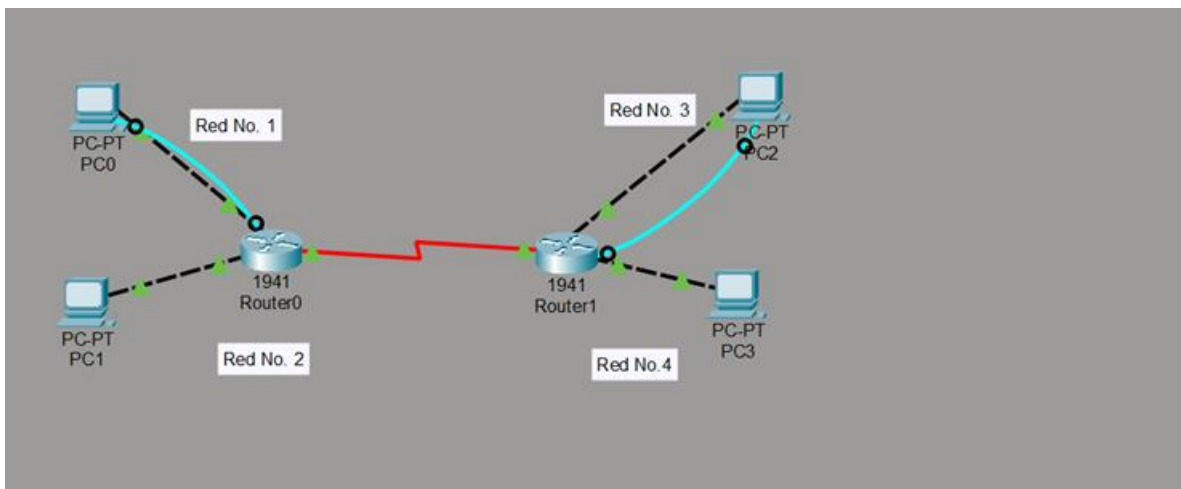
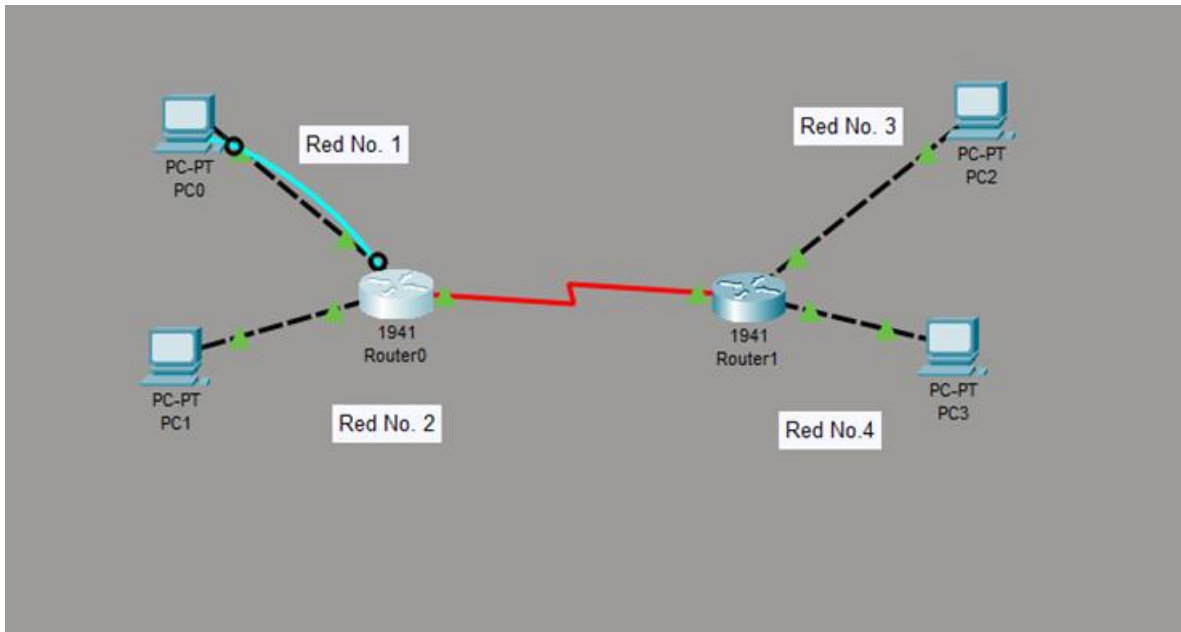


1. Ubicamos los pc y routers como indican la imagen

2. Apagamos los router 1 y 2 y agregamos el HWIC-2T y el HWIC-4ESW a cada router y volvemos a prender



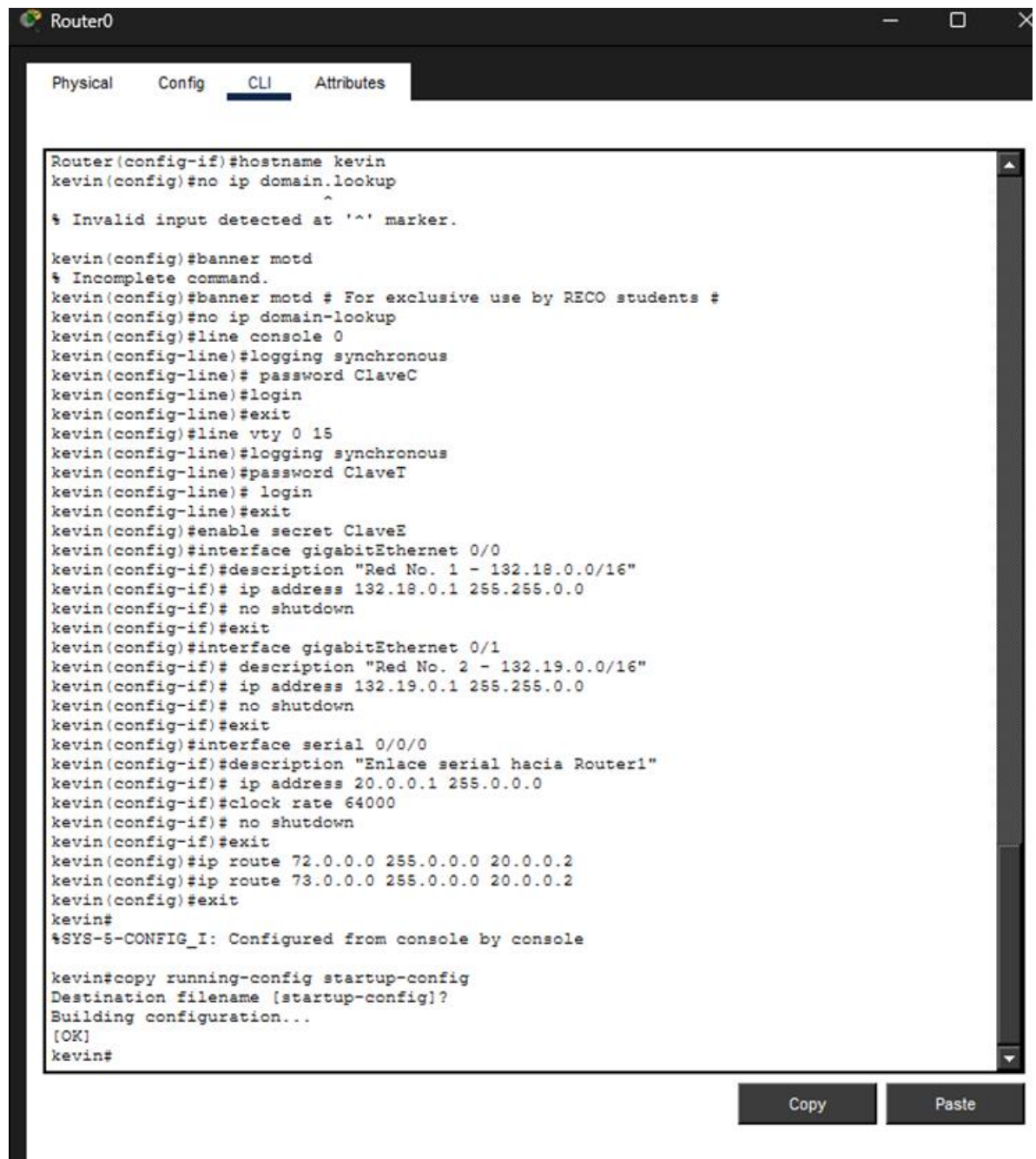
3. Conectamos el router 0 con el pc 0 el cable azul a console y al RS232 al pc 0



Kevin angel

4. Configuramos cada router el gigabitEthernet 0/0, gigabitEthernet 0/1 y serial 0/0/0 y cada pc además de su ip asignada

Router 0



The screenshot shows a Cisco Router CLI window titled "Router0". The interface has tabs for "Physical", "Config", "CLI", and "Attributes", with "CLI" currently selected. The command history shows the following sequence of commands and their outputs:

```
Router(config-if)#hostname kevin
kevin(config)#no ip domain.lookup
^
% Invalid input detected at '^' marker.

kevin(config)#banner motd
% Incomplete command.
kevin(config)#banner motd # For exclusive use by RECO students #
kevin(config)#no ip domain-lookup
kevin(config)#line console 0
kevin(config-line)#logging synchronous
kevin(config-line)# password ClaveC
kevin(config-line)#login
kevin(config-line)#exit
kevin(config)#line vty 0 15
kevin(config-line)#logging synchronous
kevin(config-line)#password ClaveT
kevin(config-line)# login
kevin(config-line)#exit
kevin(config)#enable secret ClaveE
kevin(config)#interface gigabitEthernet 0/0
kevin(config-if)#description "Red No. 1 - 132.18.0.0/16"
kevin(config-if)# ip address 132.18.0.1 255.255.0.0
kevin(config-if)# no shutdown
kevin(config-if)#exit
kevin(config)#interface gigabitEthernet 0/1
kevin(config-if)# description "Red No. 2 - 132.19.0.0/16"
kevin(config-if)# ip address 132.19.0.1 255.255.0.0
kevin(config-if)# no shutdown
kevin(config-if)#exit
kevin(config)#interface serial 0/0/0
kevin(config-if)#description "Enlace serial hacia Router1"
kevin(config-if)# ip address 20.0.0.1 255.0.0.0
kevin(config-if)#clock rate 64000
kevin(config-if)# no shutdown
kevin(config-if)#exit
kevin(config)#ip route 72.0.0.0 255.0.0.0 20.0.0.2
kevin(config)#ip route 73.0.0.0 255.0.0.0 20.0.0.2
kevin(config)#exit
kevin#
%SYS-5-CONFIG_I: Configured from console by console

kevin#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
kevin#
```

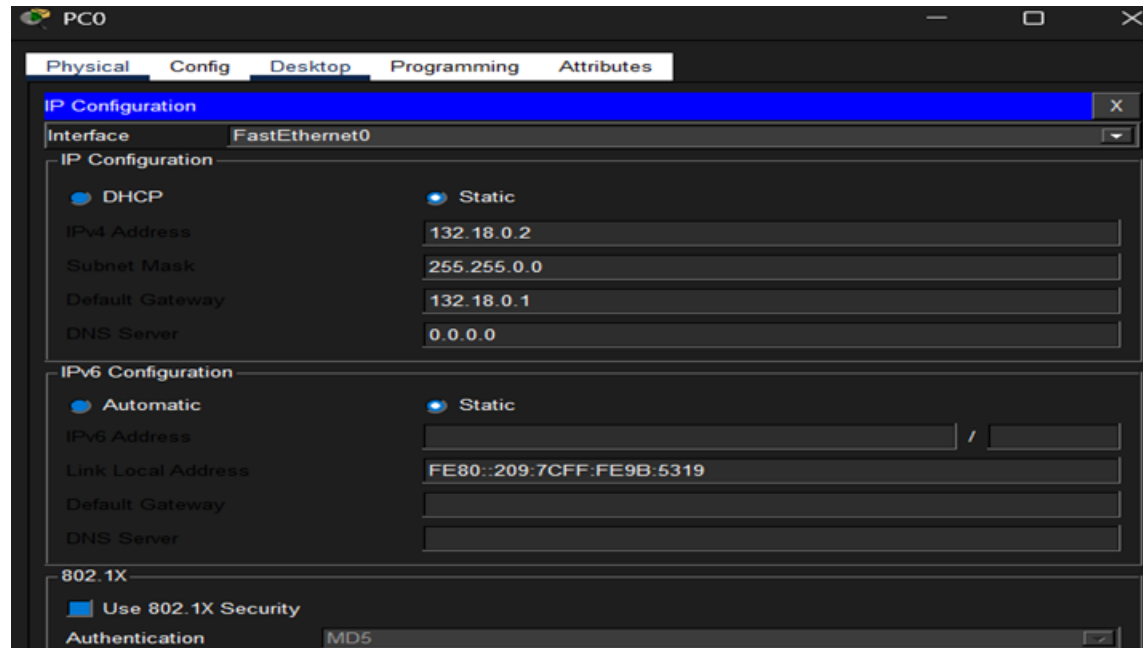
At the bottom right of the window, there are two buttons: "Copy" and "Paste".

Router 1

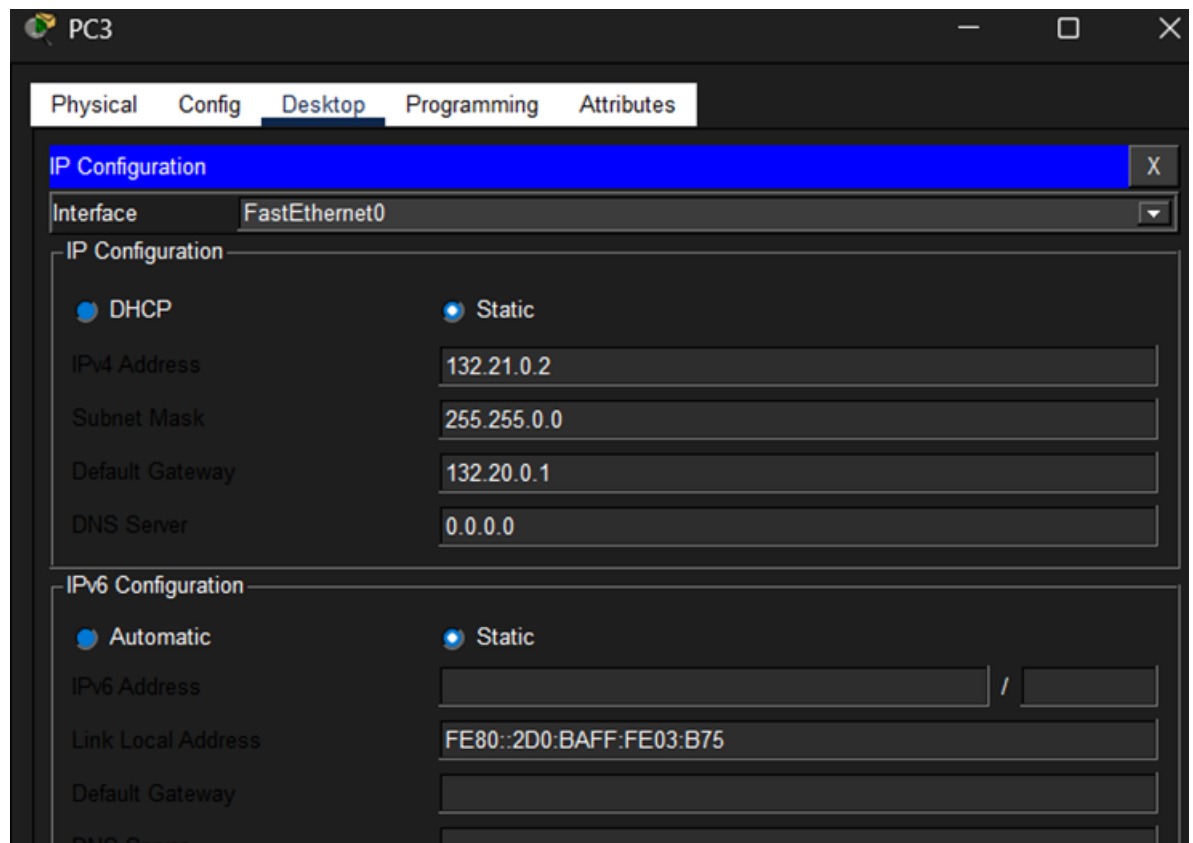
PRESS RETURN TO GET STARTED:

```
Router>enable
Router#
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0
Router(config-if)#ip address 132.20.0.1 255.255.0.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/1
Router(config-if)#ip address 132.21.0.1 255.255.0.0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#ip address 20.0.0.2 255.0.0.0
Router(config-if)#
```

Pc 0







Probamos la conectividad

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 132.19.0.2

Pinging 132.19.0.2 with 32 bytes of data:

Request timed out.
Reply from 132.19.0.2: bytes=32 time<1ms TTL=127
Reply from 132.19.0.2: bytes=32 time<1ms TTL=127
Reply from 132.19.0.2: bytes=32 time<1ms TTL=127

Ping statistics for 132.19.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 132.20.0.2

Pinging 132.20.0.2 with 32 bytes of data:

Reply from 132.18.0.1: Destination host unreachable.
Reply from 132.18.0.1: Destination host unreachable.
Reply from 132.18.0.1: Destination host unreachable.
Reply from 132.18.0.1: Destination host unreachable.

Ping statistics for 132.20.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 132.20.0.2

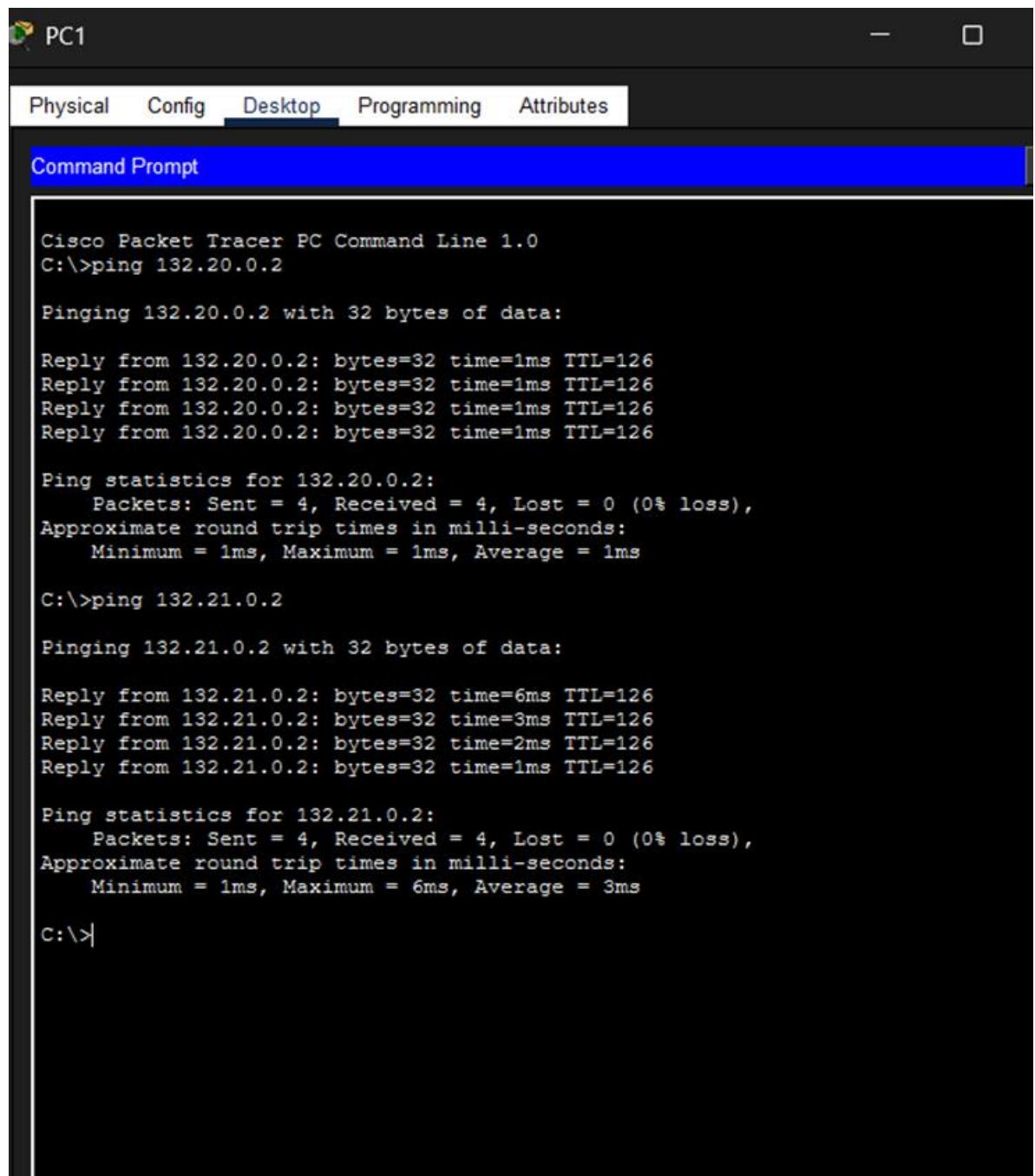
Pinging 132.20.0.2 with 32 bytes of data:

Request timed out.
Reply from 132.20.0.2: bytes=32 time=2ms TTL=126
Reply from 132.20.0.2: bytes=32 time=1ms TTL=126
Reply from 132.20.0.2: bytes=32 time=1ms TTL=126

Ping statistics for 132.20.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>|
```

Top



```
PC1
Physical Config Desktop Programming Attributes
Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 132.20.0.2

Pinging 132.20.0.2 with 32 bytes of data:

Reply from 132.20.0.2: bytes=32 time=1ms TTL=126
Reply from 132.20.0.2: bytes=32 time=1ms TTL=126
Reply from 132.20.0.2: bytes=32 time=1ms TTL=126
Reply from 132.20.0.2: bytes=32 time=1ms TTL=126

Ping statistics for 132.20.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\>ping 132.21.0.2

Pinging 132.21.0.2 with 32 bytes of data:

Reply from 132.21.0.2: bytes=32 time=6ms TTL=126
Reply from 132.21.0.2: bytes=32 time=3ms TTL=126
Reply from 132.21.0.2: bytes=32 time=2ms TTL=126
Reply from 132.21.0.2: bytes=32 time=1ms TTL=126

Ping statistics for 132.21.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 6ms, Average = 3ms

C:\>
```

Probamos el tracert

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt

Ping statistics for 132.20.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 132.20.0.2

Pinging 132.20.0.2 with 32 bytes of data:

Request timed out.
Reply from 132.20.0.2: bytes=32 time=2ms TTL=126
Reply from 132.20.0.2: bytes=32 time=1ms TTL=126
Reply from 132.20.0.2: bytes=32 time=1ms TTL=126

Ping statistics for 132.20.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>ping 132.21.0.2

Pinging 132.21.0.2 with 32 bytes of data:

Request timed out.
Reply from 132.21.0.2: bytes=32 time=1ms TTL=126
Reply from 132.21.0.2: bytes=32 time=1ms TTL=126
Reply from 132.21.0.2: bytes=32 time=2ms TTL=126

Ping statistics for 132.21.0.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\>tracert 132.20.0.2

Tracing route to 132.20.0.2 over a maximum of 30 hops:

  0  0 ms    0 ms    0 ms    132.18.0.1
  1  0 ms    3 ms    1 ms    20.0.0.2
  2  0 ms    1 ms    1 ms    132.20.0.2

Trace complete.

C:\>
```

```
Madero(config)#hostname Madero
Madero(config)#interface GigabitEthernet0/0
Madero(config-if)#ip address 74.0.0.1 255.0.0.0
Madero(config-if)#no shutdown
Madero(config-if)#exit
Madero(config)#interface GigabitEthernet0/1
Madero(config-if)#ip address 75.0.0.1 255.0.0.0
Madero(config-if)#no shutdown
Madero(config-if)#exit
Madero(config)#interface serial0/0/0
Madero(config-if)#ip address 20.0.0.2 255.0.0.0
Madero(config-if)#no shutdown
Madero(config-if)#exit
Madero(config)#no ip route 132.18.0.0 255.255.0.0 serial0/0/0
Madero(config)#no ip route 132.19.0.0 255.255.0.0 serial0/0/0
Madero(config)#ip route 72.0.0.0 255.0.0.0 serial0/0/0
%Default route without gateway, if not a point-to-point interface, may impact
performance
Madero(config)#ip route 73.0.0.0 255.0.0.0 serial0/0/0
%Default route without gateway, if not a point-to-point interface, may impact
performance
Madero(config)#end
Madero#
%SYS-5-CONFIG_I: Configured from console by console

Madero#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Madero#
```

```
JhonatanNembre(config)#hostname Jhonatan
Jhonatan(config)#interface GigabitEthernet0/0
Jhonatan(config-if)#ip address 72.0.0.1 255.0.0.0
Jhonatan(config-if)#no shutdown
Jhonatan(config-if)#exit
Jhonatan(config)#interface GigabitEthernet0/1
Jhonatan(config-if)#ip address 73.0.0.1 255.0.0.0
Jhonatan(config-if)#no shutdown
Jhonatan(config-if)#exit
Jhonatan(config)#interface serial0/0/0
Jhonatan(config-if)#ip address 20.0.0.1 255.0.0.0
Jhonatan(config-if)#clock rate 64000
Jhonatan(config-if)#no shutdown
Jhonatan(config-if)#exit
Jhonatan(config)#no ip route 132.20.0.0 255.255.0.0 serial0/0/0
Jhonatan(config)#no ip route 132.21.0.0 255.255.0.0 serial0/0/0
Jhonatan(config)#ip route 74.0.0.0 255.0.0.0 serial0/0/0
%Default route without gateway, if not a point-to-point interface, may impact
performance
Jhonatan(config)#ip route 75.0.0.0 255.0.0.0 serial0/0/0
%Default route without gateway, if not a point-to-point interface, may impact
performance
Jhonatan(config)#end
Jhonatan#
%SYS-5-CONFIG_I: Configured from console by console

Jhonatan#copy running-config startup-config
Destination filename [startup-config]? y
%Error copying nvram:y (Invalid argument)
Jhonatan#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Jhonatan#
```

PC0

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

72.0.0.2

Subnet Mask

255.255.0.0

Default Gateway

72.0.0.1

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::209:7CFF:FE9B:5319

Default Gateway

DNS Server

802.1X

PC1

Physical

Config

Desktop

Programming

Attributes

IP Configuration

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

73.0.0.2

Subnet Mask

255.255.0.0

Default Gateway

73.0.0.1

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::201:64FF:FE51:EDE6

Default Gateway

DNS Server

802.1X

Use 802.1X Security

Authentication

MD5

Username

Password

PC2

Physical

Config

Desktop

Programming

Attributes

IP Configuration

X

Interface

FastEthernet0

IP Configuration

DHCP

Static

IPv4 Address

74.0.0.2

Subnet Mask

255.0.0.0

Default Gateway

74.0.0.1

DNS Server

0.0.0.0

IPv6 Configuration

Automatic

Static

IPv6 Address

/

Link Local Address

FE80::2D0:58FF:FED3:451D

Default Gateway

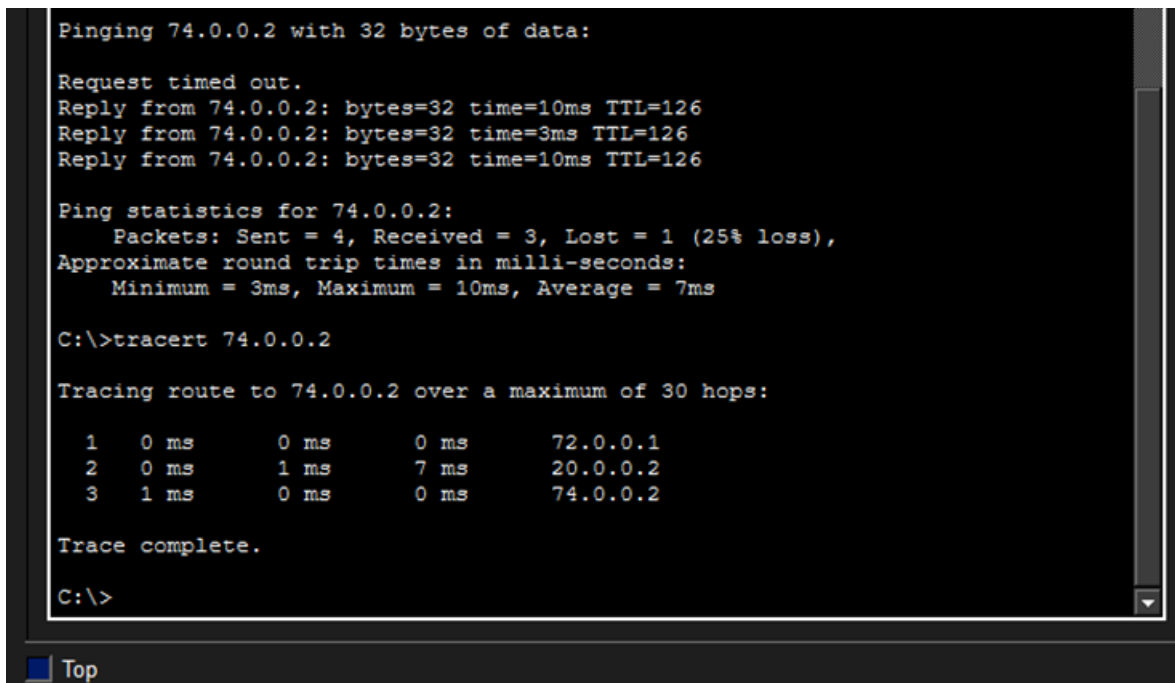
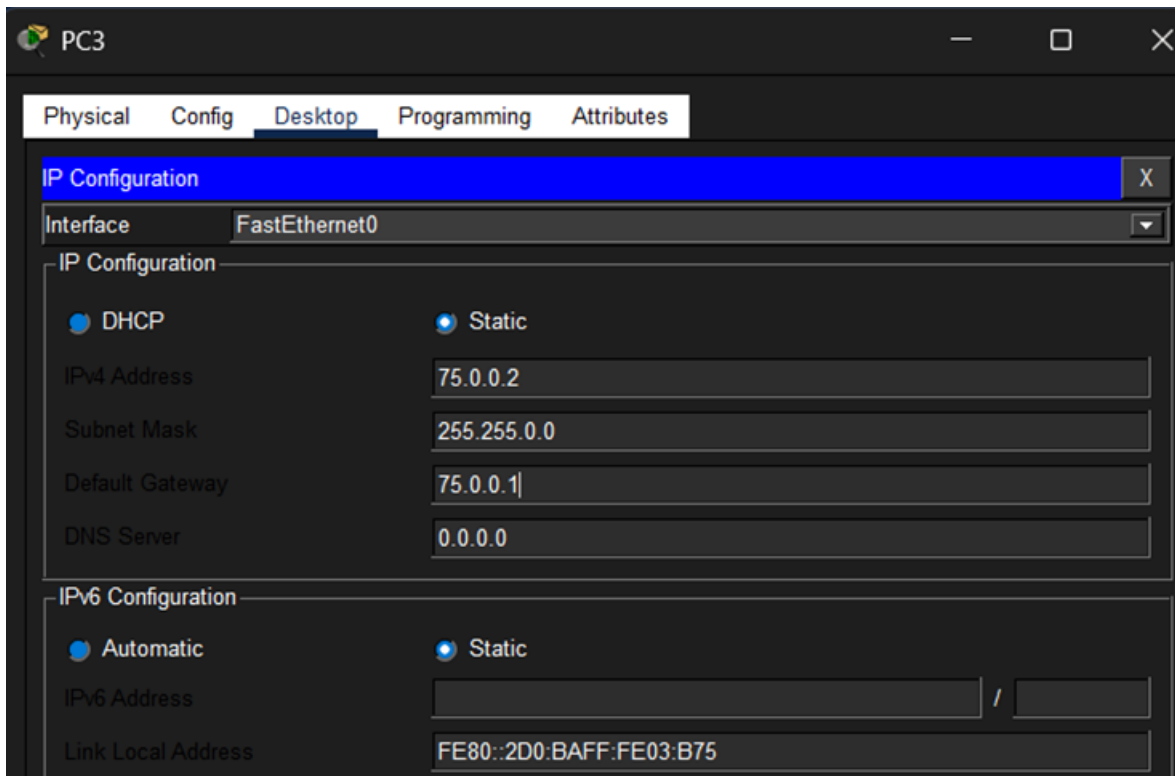
DNS Server

802.1X

Use 802.1X Security

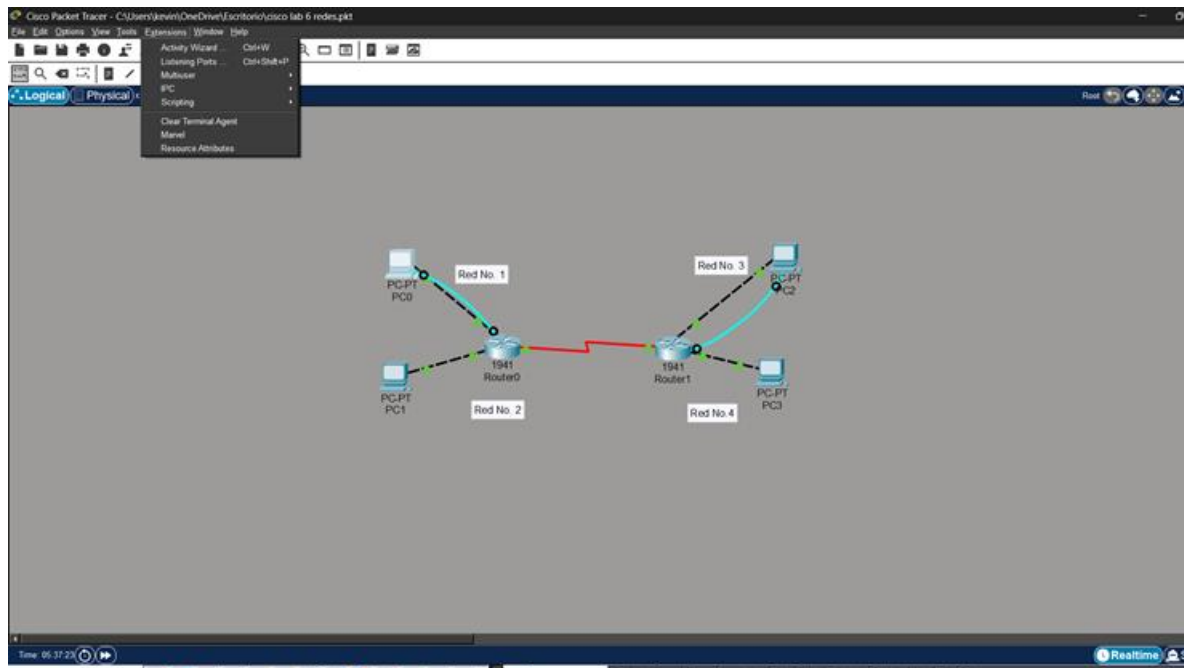
Authentication

MD5



CONEXIÓN ENTRE CISCOS

Ingresamos a extensiones multiuser en el cisco de kevin



Clave : kevin

 Multiuser Listen ✕

Local Listening Address:

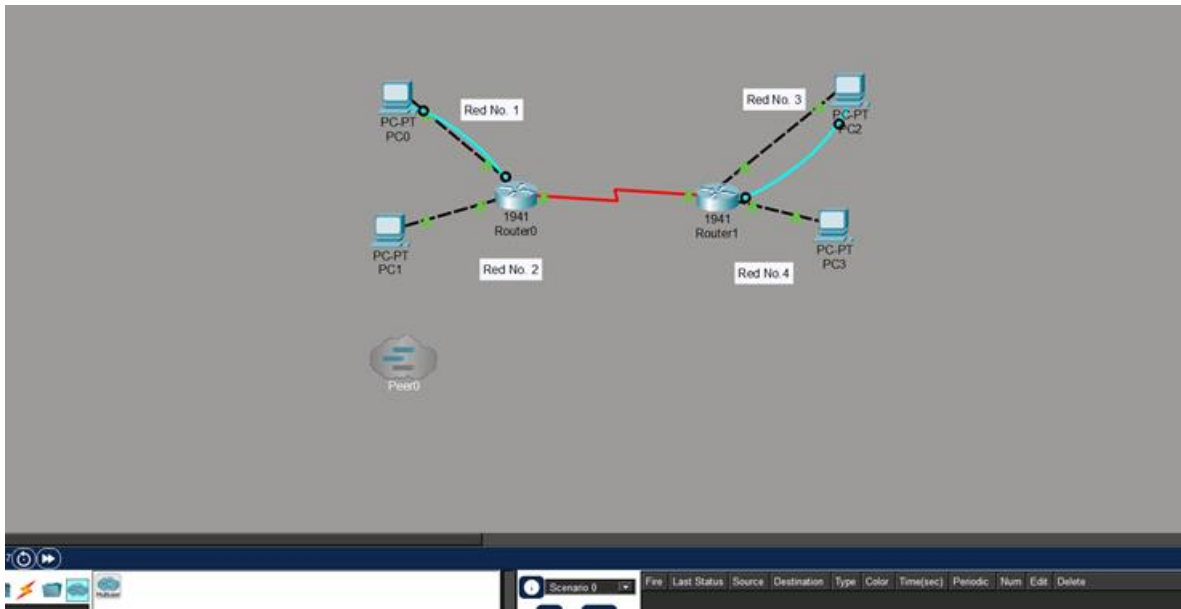
```
192.168.56.1:38000
192.168.136.1:38000
192.168.163.1:38000
192.168.20.31:38000
127.0.0.1:38000
```

Port Number:

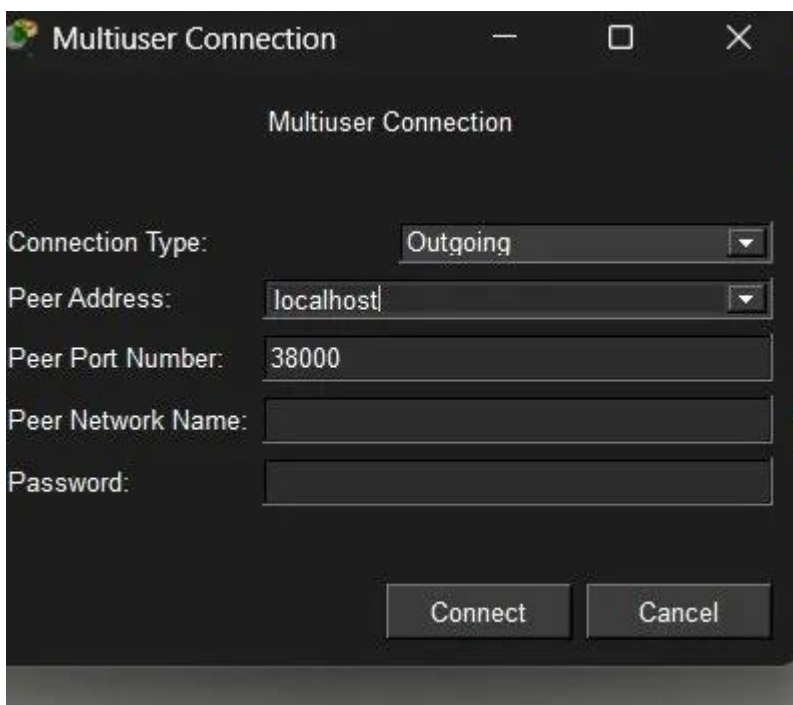
Password:

Existing Remote Networks	New Remote Networks
☐ Always Accept	☐ Always Accept
☐ Always Deny	☐ Always Deny
☒ Prompt	☒ Prompt

Damos ok y en el cisco de jhonathan arrastramos la nube de multiuser

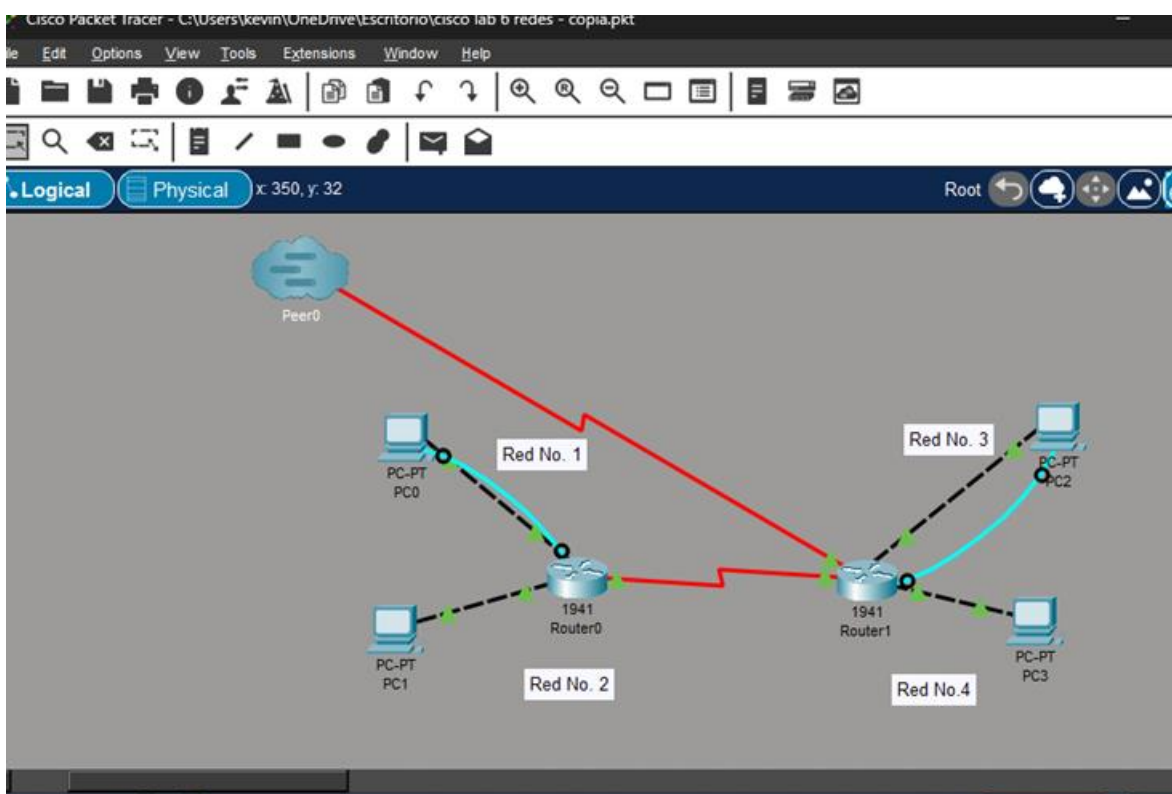
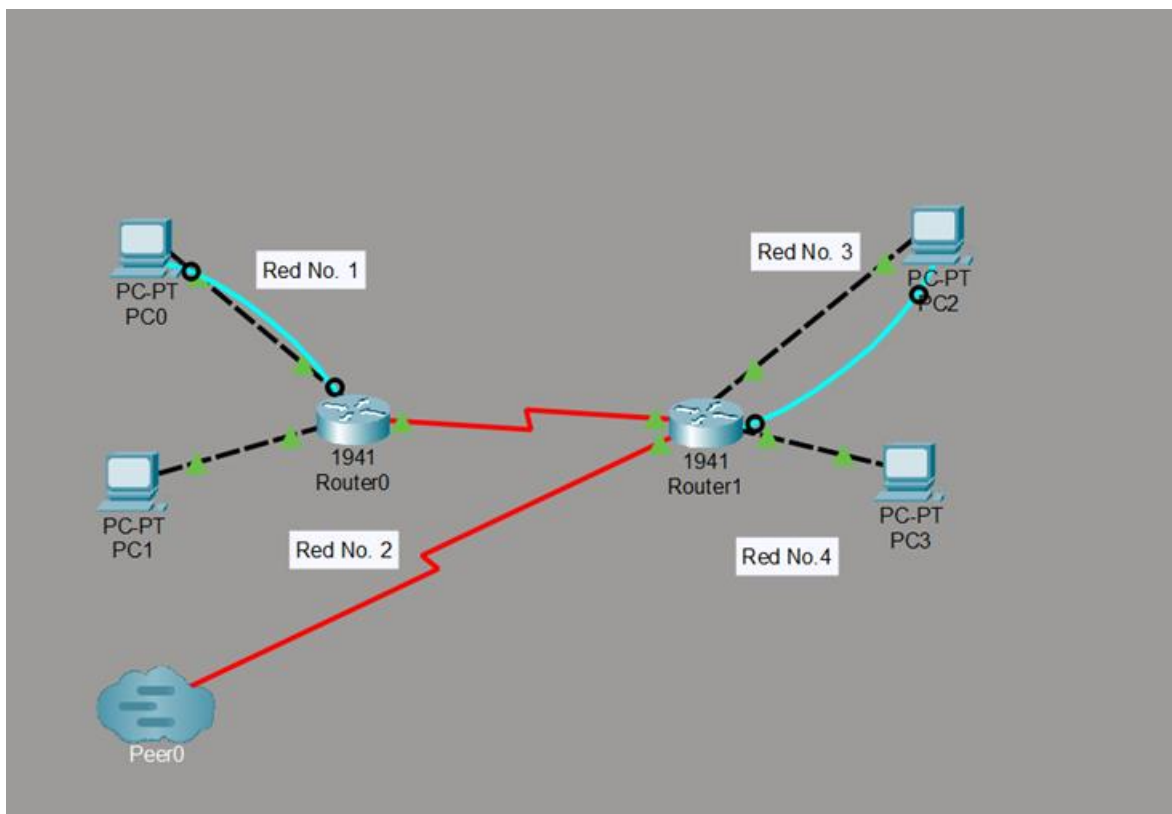


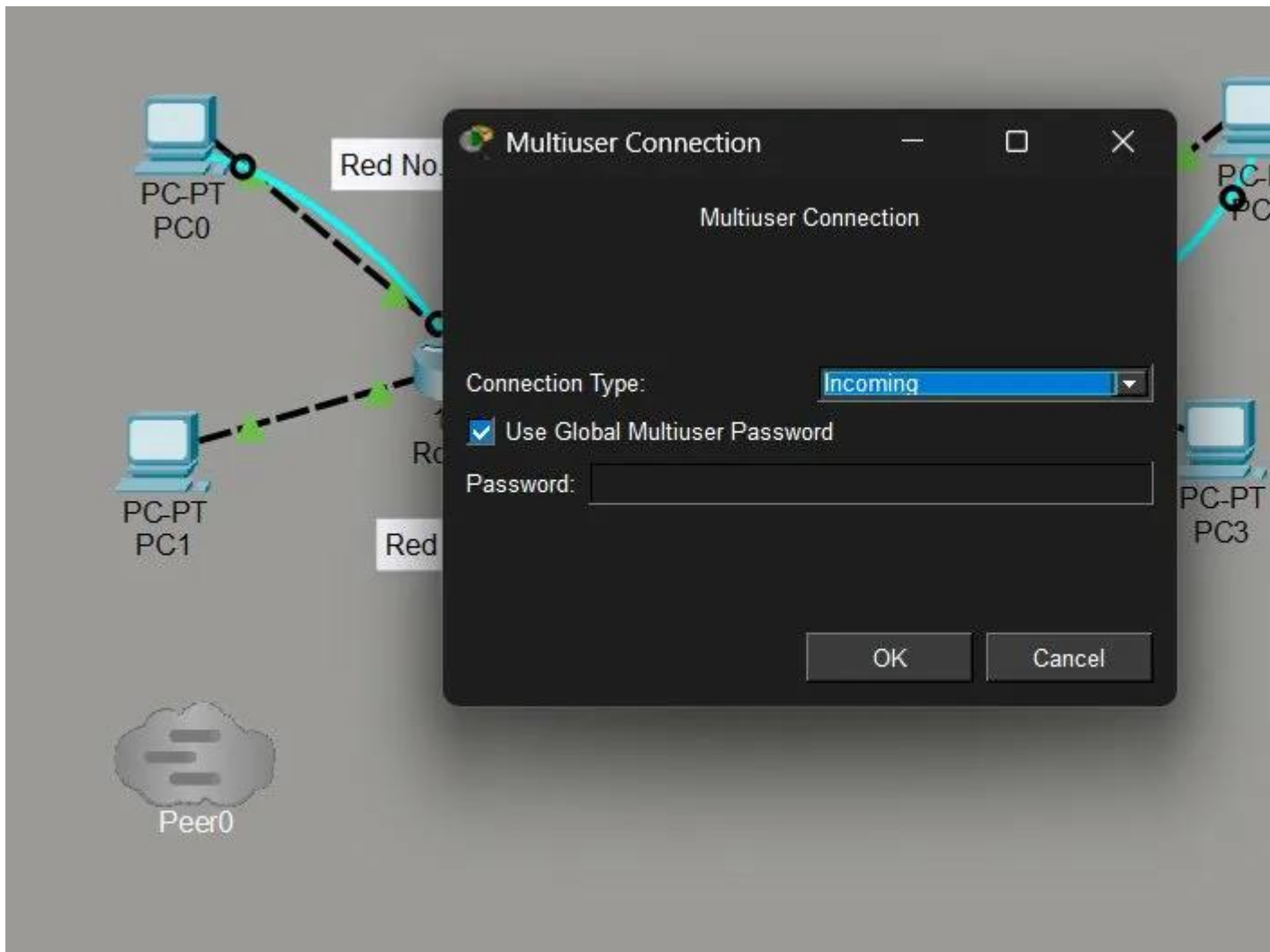
Ingresamos a ella y sale donde cambiamos el tipo de conexión a outgoing y agregamos la clave de kevin



Damos yes

Conectamos el cable serial DCE de la nube a el router 1 en ambos ciscos





Configuramos la interfaz serial

```
Password:

Angel>enable
Password:
Password:
Angel#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Angel(config)#interface serial0/0/1
Angel(config-if)#ip address 30.0.0.1 255.0.0.0
Angel(config-if)#clock rate 64000
Angel(config-if)#no shutdown
Angel(config-if)#exit
Angel(config)#end
Angel#
%SYS-5-CONFIG_I: Configured from console by console

Angel#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Angel#
```

 [Top](#)

```
Madero#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Madero#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Madero(config)#interface serial0/0/1
Madero(config-if)#no ip address 20.0.0.2 255.0.0.0
Madero(config-if)#ip address 30.0.0.2 255.0.0.0
Madero(config-if)#no shutdown
Madero(config-if)#exit
Madero(config)#end
Madero#
%SYS-5-CONFIG_I: Configured from console by console

Madero#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Madero#
```

 [Top](#)

Probamos que sirva

For exclusive use by RECO students

User Access Verification

Password:

Jhonatan>enable

Password:

Password:

Password:

Jhonatan#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

Jhonatan(config)#ip route 132.18.0.0 255.255.0.0 serial0/0/0

%Default route without gateway, if not a point-to-point interface, may impact performance

Jhonatan(config)#ip route 132.19.0.0 255.255.0.0 serial0/0/0

%Default route without gateway, if not a point-to-point interface, may impact performance

Jhonatan(config)#ip route 132.20.0.0 255.255.0.0 serial0/0/0

%Default route without gateway, if not a point-to-point interface, may impact performance

Jhonatan(config)#ip route 132.21.0.0 255.255.0.0 serial0/0/0

%Default route without gateway, if not a point-to-point interface, may impact performance

Jhonatan(config)#end

Jhonatan#

%SYS-5-CONFIG_I: Configured from console by console

Jhonatan#copy running-config startup-config

Destination filename [startup-config]?

Building configuration...

[OK]

Jhonatan#

Top

```

If you require further assistance please contact us by sending email to
export@cisco.com.

Cisco CISC01941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
4 FastEthernet interface(s)
2 Gigabit Ethernet interfaces
2 Low-speed serial(sync/async) network interface(s)
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)

Press RETURN to get started!

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up

For exclusive use by RECO students

User Access Verification

Password:

kevin>enable
Password:
kevin#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
kevin(config)#ip route 72.0.0.0 255.0.0.0 serial0/0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
kevin(config)#ip route 73.0.0.0 255.0.0.0 serial0/0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
kevin(config)#ip route 74.0.0.0 255.0.0.0 serial0/0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
kevin(config)#ip route 75.0.0.0 255.0.0.0 serial0/0/0
%Default route without gateway, if not a point-to-point interface, may impact performance
kevin(config)#end
kevin#
%SYS-5-CONFIG_I: Configured from console by console

kevin#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
kevin#

```

Base Software Installation

Part of the foundational platform of an organization's IT infrastructure involves web services, which can be hosted within the company's data center or on a cloud server. These services store the organization's web pages and are used by various clients. In this lab, we will implement this service.

1. Web Service Installation

[For groups of 1, 2, and 3 students]

- a. Install the Apache web server on the virtual machine with the FreeBSD operating system.

- b. Install the Nginx web server on the virtual machine with the Slackware Linux operating system.
- c. Configure the web server that comes with Windows Server.
- d. Create a simple webpage on each server to test that it works.
- e. Configure the web servers to start automatically when the operating system boots.
- f. From another computer, test access to the web servers.
- g. Configure the DNS service so that each web server can be accessed by name.

Instalamos apache

```
PING google.com (172.217.29.14): 56 data bytes
64 bytes from 172.217.29.14: icmp_seq=0 ttl=115 time=6.276 ms
64 bytes from 172.217.29.14: icmp_seq=1 ttl=115 time=4.785 ms
64 bytes from 172.217.29.14: icmp_seq=2 ttl=115 time=5.037 ms
64 bytes from 172.217.29.14: icmp_seq=3 ttl=115 time=6.110 ms
64 bytes from 172.217.29.14: icmp_seq=4 ttl=115 time=5.691 ms
^Z
Suspended
root@JK:~ # pkg install apache24
Updating FreeBSD repository catalogue...
Fetching data: 100% 10 MiB 2.7 M/s 00:04
Processing entries: 100%
FreeBSD repository update completed. 36835 packages processed.
All repositories are up to date.
New version of pkg detected; it needs to be installed first.
The following 1 package(s) will be affected (of 0 checked):

Installed packages to be UPGRADED:
  pkg: 2.5.1 -> 2.6.2_1

Number of packages to be upgraded: 1

5 MiB to be downloaded.

Proceed with this action? [y/N]: █
```

Le damos y para yes

```

5 MiB to be downloaded.

Proceed with this action? [y/N]: y
[1/1] Fetching pkg-2.6.2_1: 100% 5 MiB 5.5 M/s 00:01
Checking integrity... done (0 conflicting)
[1/1] Upgrading pkg from 2.5.1 to 2.6.2_1...
[1/1] Extracting pkg-2.6.2_1: 100%
Updating FreeBSD repository catalogue...
FreeBSD repository is up to date.
All repositories are up to date.
The following 5 package(s) will be affected (of 0 checked):

New packages to be INSTALLED:
  apache24: 2.4.66
  apr: 1.7.6.1.6.3
  curl: 8.19.0_1
  libpsl: 0.21.5_2
  wolfssl: 5.9.0

Number of packages to be installed: 5

The process will require 43 MiB more space.
9 MiB to be downloaded.

Proceed with this action? [y/N]: █

```

También yes

```

configurations to use the added ODBC driver and FreeTDS' ODBC features.
=====
Message from apache24-2.4.66:

--
To run apache www server from startup, add apache24_enable="yes"
in your /etc/rc.conf. Extra options can be found in startup script.

Your hostname must be resolvable using at least 1 mechanism in
/etc/nsswitch.conf typically DNS or /etc/hosts or apache might
have issues starting depending on the modules you are using.

- apache24 default build changed from static MPM to modular MPM
- more modules are now enabled per default in the port
- icons and error pages moved from WWWDIR to DATADIR

If built with modular MPM and no MPM is activated in
httpd.conf, then mpm_prefork will be activated as default
MPM in etc/apache24/modules.d to keep compatibility with
existing php/perl/python modules!

Please compare the existing httpd.conf with httpd.conf.sample
and merge missing modules/instructions into httpd.conf!
root@JK:~ # █

```

Para que se inicie automáticamente cuando se inicie el sistema sysrc
apache24_enable="YES"

```
sysrc: unknown variable 'apache24_eneb'  
root@JK:~ # sysrc apache24_enable="YES"  
apache24_enable: -> YES  
root@JK:~ # █
```

Lo iniciamos con service apache24 start

```
root@JK:~ # service apache24 start  
Performing sanity check on apache24 configuration:  
AH00557: httpd: apr_sockaddr_info_get() failed for JK  
AH00558: httpd: Could not reliably determine the server's fully qualified domain  
name, using 127.0.0.1. Set the 'ServerName' directive globally to suppress this  
message  
Syntax OK  
Starting apache24.  
AH00557: httpd: apr_sockaddr_info_get() failed for JK  
AH00558: httpd: Could not reliably determine the server's fully qualified domain  
name, using 127.0.0.1. Set the 'ServerName' directive globally to suppress this  
message  
root@JK:~ # █
```

creamos una página web simple:

```
tee /usr/local/www/apache24/data/index.html << EOF
```

```
<h1>Hola desde FreeBSD - Student 01</h1>
```

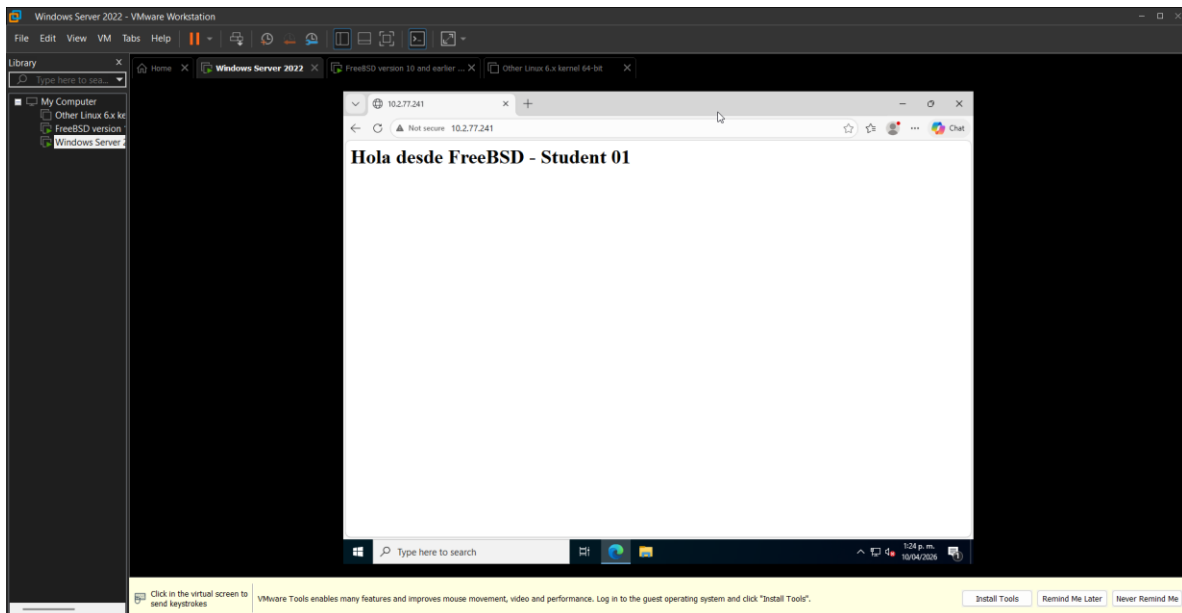
```
EOF
```

```
root@JK:~ # tee /usr/local/www/apache24/data/index.html << EOF  
? <h1>Hola desde FreeBSD - Student 01</h1>  
? EOF  
<h1>Hola desde FreeBSD - Student 01</h1>  
root@JK:~ # |
```

```
ifconfig | grep inet
```

```
root@JK:~ # ifconfig | grep inet
    inet 10.2.77.241 netmask 0xffff0000 broadcast 10.2.255.255
    inet6 ::1 prefixlen 128
    inet6 fe80::1%lo0 prefixlen 64 scopeid 0x2
    inet 127.0.0.1 netmask 0xff000000
root@JK:~ # |
```

Desde Windows Server abrimos el navegador (Internet Explorer) y escribimos en la barra de direcciones:



instalamos sbopkg que es el gestor de SlackBuilds:

De un repositorio de github instalamos con wget

https://github.com/sbopkg/sbopkg/releases/download/0.38.2/sbopkg-0.38.2-noarch-1_wsr.tgz

```

HTTP request sent, awaiting response... 302 Found
Location: https://release-assets.githubusercontent.com/github-production-release-asset/37267968/2783622c-18c8-4e2d-a7b3-f797d7f44c48?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-04-10T21%3A07%3A35Z&rsd=attachment%3B+filename%3Dsbopkg-0.38.2-noarch-1_wsr.tgz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-04-10T20%3A07%3A04Z&ske=2026-04-10T21%3A07%3A35Z&sks=b&skv=2018-11-09&sig=IL5fLo7Z2o510tU4Q8jlsmf28CTjaaX0oaJSHXC%2FLMM%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3NTg1MzU1OSwibmJmIjoxNzc1ODUzMjU5LCJwYXRoIjoicmVsZWZzZWZc2V0cHJvZHVjdGlubi5ibG9iLmNvcuUud2luZG93cy5uZXQifQ.MAX9bLTL6KQq1rERDP8i4qeiBuKWLLtl1el3XSGsDxM&response-content-disposition=attachment%3B%20filename%3Dsbopkg-0.38.2-noarch-1_wsr.tgz&response-content-type=application%2Foctet-stream [following]
--2026-04-10 15:34:19-- https://release-assets.githubusercontent.com/github-production-release-asset/37267968/2783622c-18c8-4e2d-a7b3-f797d7f44c48?sp=r&sv=2018-11-09&sr=b&spr=https&se=2026-04-10T21%3A07%3A35Z&rsd=attachment%3B+filename%3Dsbopkg-0.38.2-noarch-1_wsr.tgz&rsct=application%2Foctet-stream&skoid=96c2d410-5711-43a1-aedd-ab1947aa7ab0&sktid=398a6654-997b-47e9-b12b-9515b896b4de&skt=2026-04-10T20%3A07%3A04Z&ske=2026-04-10T21%3A07%3A35Z&sks=b&skv=2018-11-09&sig=IL5fLo7Z2o510tU4Q8jlsmf28CTjaaX0oaJSHXC%2FLMM%3D&jwt=eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJnaXRodWIuY29tIiwia2V5Ijoia2V5MSIsImV4cCI6MTc3NTg1MzU1OSwibmJmIjoxNzc1ODUzMjU5LCJwYXRoIjoicmVsZWZzZWZc2V0cHJvZHVjdGlubi5ibG9iLmNvcuUud2luZG93cy5uZXQifQ.MAX9bLTL6KQq1rERDP8i4qeiBuKWLLtl1el3XSGsDxM&response-content-disposition=attachment%3B%20filename%3Dsbopkg-0.38.2-noarch-1_wsr.tgz&response-content-type=application%2Foctet-stream
Resolving release-assets.githubusercontent.com (release-assets.githubusercontent.com)... 185.199.111.133, 185.199.110.133, 185.199.108.133, ...
Connecting to release-assets.githubusercontent.com (release-assets.githubusercontent.com)|185.199.111.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 141891 (139K) [application/octet-stream]
Saving to: 'sbopkg-0.38.2-noarch-1_wsr.tgz.1'

sbopkg-0.38.2-noar 100%[=====>] 138.57K  --.-KB/s    in 0.03s

2026-04-10 15:34:19 (4.01 MB/s) - 'sbopkg-0.38.2-noarch-1_wsr.tgz.1' saved [141891/141891]

root@jk:~# |

```

Ahora instala el paquete:

```
installpkg sbopkg-0.38.2-noarch-1_wsr.tgz
```

```

root@jk:~# installpkg sbopkg-0.38.2-noarch-1_wsr.tgz
Verifying package sbopkg-0.38.2-noarch-1_wsr.tgz.
Installing package sbopkg-0.38.2-noarch-1_wsr.tgz:
PACKAGE DESCRIPTION:
# sbopkg (SlackBuilds.org Package Browser)
#
# Sbopkg is a command-line and dialog-based tool to interact with the
# SlackBuilds.org repository, a collection of third-party SlackBuild
# scripts to build Slackware packages.
#
# Homepage : https://www.sbopkg.org
# Development: https://github.com/sbopkg/sbopkg
#
Executing install script for sbopkg-0.38.2-noarch-1_wsr.tgz.
Package sbopkg-0.38.2-noarch-1_wsr.tgz installed.
root@jk:~# |

```

```

system/sbozyp/
system/sbozyp/sbozyp.SlackBuild
system/sbozyp/sbozyp.info
system/sdate/
system/sdate/README
system/sdate/sdate.1.diff
system/sdate/sdate.SlackBuild
system/sdate/sdate.info
system/sdate/slack-desc
system/slpkg/
system/slpkg/slpkg.SlackBuild
system/slpkg/slpkg.info
system/torrent-file-editor/
system/torrent-file-editor/torrent-file-editor.SlackBuild
system/torrent-file-editor/torrent-file-editor.info
system/usermin/
system/usermin/usermin.SlackBuild
system/usermin/usermin.info
system/webmin/
system/webmin/webmin.SlackBuild
system/webmin/webmin.info
system/xfile/
system/xfile/xfile.SlackBuild
system/xfile/xfile.info
system/yq/
system/yq/yq.SlackBuild
system/yq/yq.info
system/yubico-piv-tool/
system/yubico-piv-tool/yubico-piv-tool.SlackBuild
system/yubico-piv-tool/yubico-piv-tool.info

sent 123,389 bytes received 9,817,510 bytes 336,979.63 bytes/sec
total size is 145,591,152 speedup is 14.65

Rsync with the SBo repository for Slackware 15.0 complete.

Importing SBo repository for Slackware 15.0 GPG Key...
Import done.

***SYNC COMPLETE***
root@jk:~# |

```

Instalamos Nginx: sbopkg -i nginx

```
#####
      New queue process started on:
      Fri Apr 10 15:38:14 -05 2026
#####

+++++
PRE-CHECK LOG
Using the SBo repository for Slackware 15.0
Queue Process:  Download, build, and install

nginx:
  Checking GPG for nginx.tar.gz ... OK
  Processing nginx 1.28.3-1
  Using original .info file
  Using original SlackBuild file
  No build options selected.

+++++

Pre-check complete.

Do you wish to proceed based on the search results above? Packages not
found will be skipped during the process.

(P)roceed or (Q)uit?: |
```

Escibimos p y enter:

```

Built package: nginx-1.28.3-i586-1_SBo.tgz

=====
===
| Installing new package /tmp/nginx-1.28.3-i586-1_SBo.tgz
=====
===

Verifying package nginx-1.28.3-i586-1_SBo.tgz.
Installing package nginx-1.28.3-i586-1_SBo.tgz:
PACKAGE DESCRIPTION:
# nginx (http/imap/pop3 proxy)
#
# Nginx [engine x] is a high-performance HTTP server and reverse proxy,
# as well as an IMAP/POP3 proxy server.
#
# Nginx was written by Igor Sysoev.
#
# Homepage: http://nginx.net/
#
Executing install script for nginx-1.28.3-i586-1_SBo.tgz.
Package nginx-1.28.3-i586-1_SBo.tgz installed.
Done upgrading/installing package.

+++++
SUMMARY LOG
Using the SBo repository for Slackware 15.0
Queue Process: Download, build, and install

nginx:
  MD5SUM check for nginx-1.28.3.tar.gz ... OK
  Building package nginx-1.28.3-i586-1_SBo.tgz ... OK
  Installing package nginx-1.28.3-i586-1_SBo.tgz ... OK

+++++

#####
      Queue process complete!
#####

root@jk:~# |

```

Ahora activamos para que inicie automáticamente con el sistema:

```
chmod +x /etc/rc.d/rc.nginx
```

```

root@jk:~# chmod +x /etc/rc.d/rc.nginx

```

Y lo iniciamos : /etc/rc.d/rc.nginx start

```
root@jk:~# chmod +x /etc/rc.d/rc.nginx
root@jk:~# /etc/rc.d/rc.nginx start
Starting Nginx server daemon...
root@jk:~# |
```

Ahora creamos la pagina web:

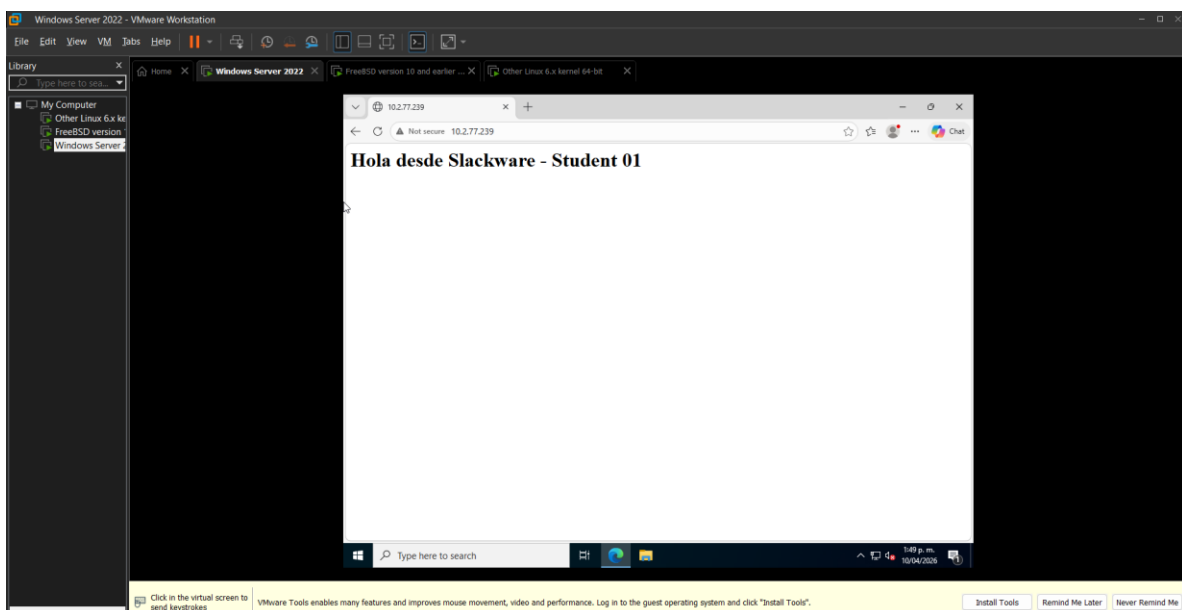
```
tee /var/www/html/index.html << EOF
```

```
<h1>Hola desde Slackware - Student 01</h1>
```

```
EOF
```

```
root@jk:~# tee /var/www/html/index.html << EOF
> <h1>Hola desde Slackware - Student 01</h1>
> EOF
<h1>Hola desde Slackware - Student 01</h1>
root@jk:~# |
```

Ahora desde Windows Server abrimos el navegador y escribimos: <http://10.2.77.239>



Listop ahora en windos :

Abrimos el Server Manager

la de la Stackoverflow Student 01

Server Manager

Dashboard

Manage Tools View Help

Dashboard

Local Server

All Servers

DNS

File and Storage Services

WELCOME TO SERVER MANAGER

1 Configure this local server

2 Add roles and features

3 Add other servers to manage

4 Create a server group

5 Connect this server to cloud services

Hide

ROLES AND SERVER GROUPS

Roles: 2 | Server groups: 1 | Servers total: 1

10.2.77.239

Add Roles and Features Wizard

Before you begin

DESTINATION SERVER
WIN-MMRM9RPM11P

Before You Begin

Installation Type

Server Selection

Server Roles

Features

Confirmation

Results

This wizard helps you install roles, role services, or features. You determine which roles, role services, or features to install based on the computing needs of your organization, such as sharing documents, or hosting a website.

To remove roles, role services, or features:
[Start the Remove Roles and Features Wizard](#)

Before you continue, verify that the following tasks have been completed:

- The Administrator account has a strong password
- Network settings, such as static IP addresses, are configured
- The most current security updates from Windows Update are installed

If you must verify that any of the preceding prerequisites have been completed, close the wizard, complete the steps, and then run the wizard again.

To continue, click Next.

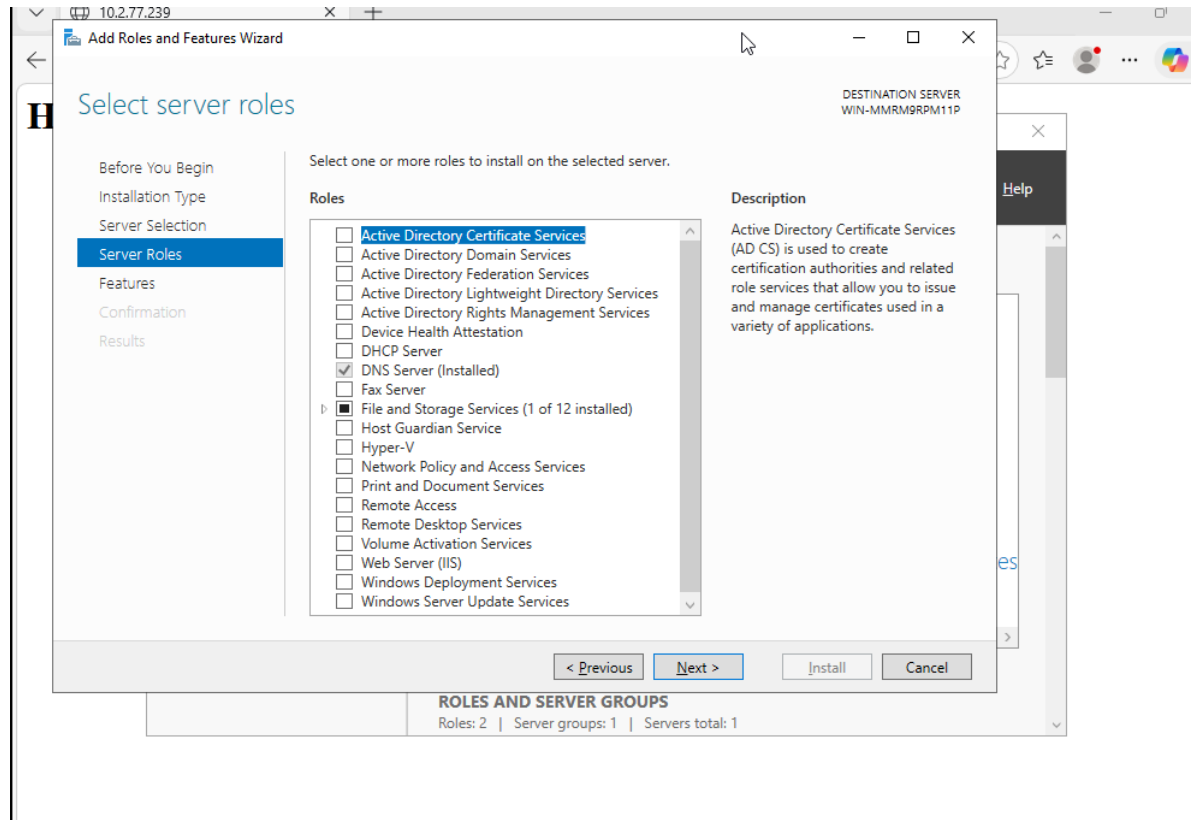
☐ Skip this page by default

< Previous Next > Install Cancel

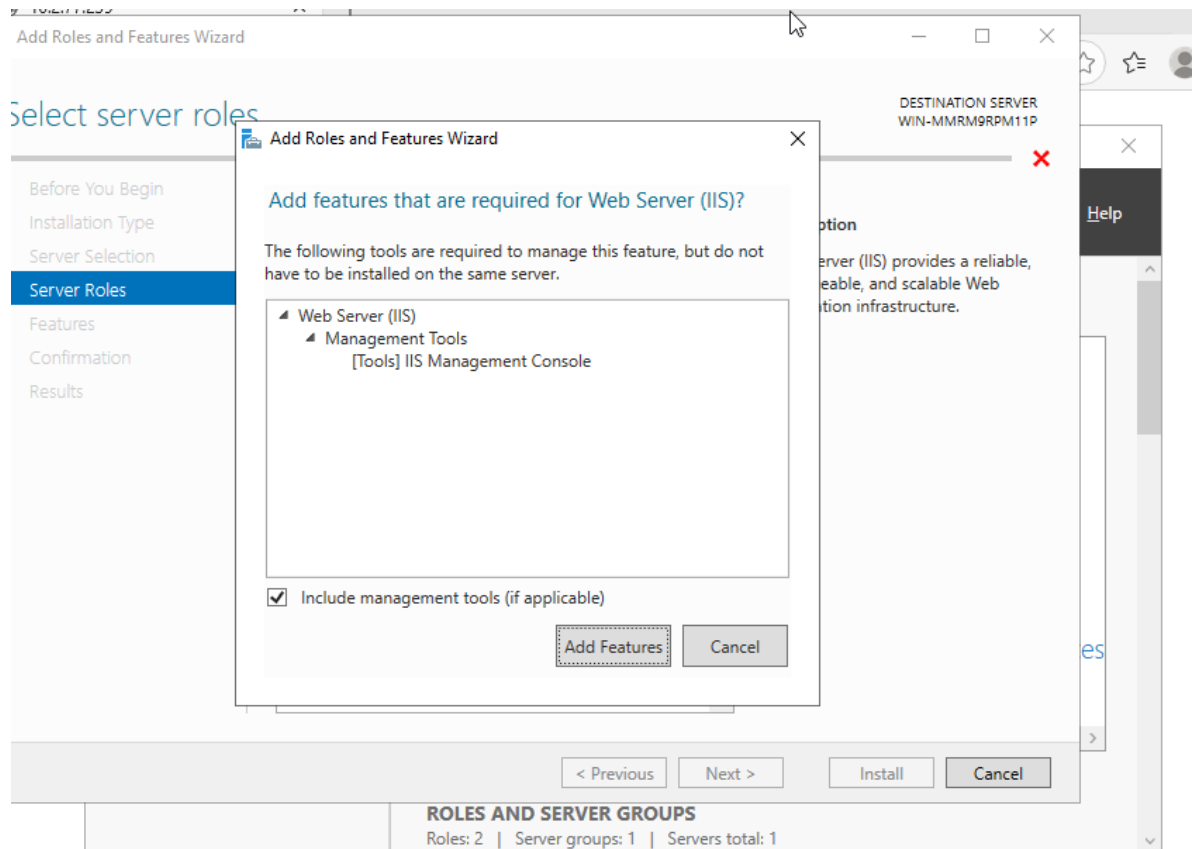
ROLES AND SERVER GROUPS

Roles: 2 | Server groups: 1 | Servers total: 1

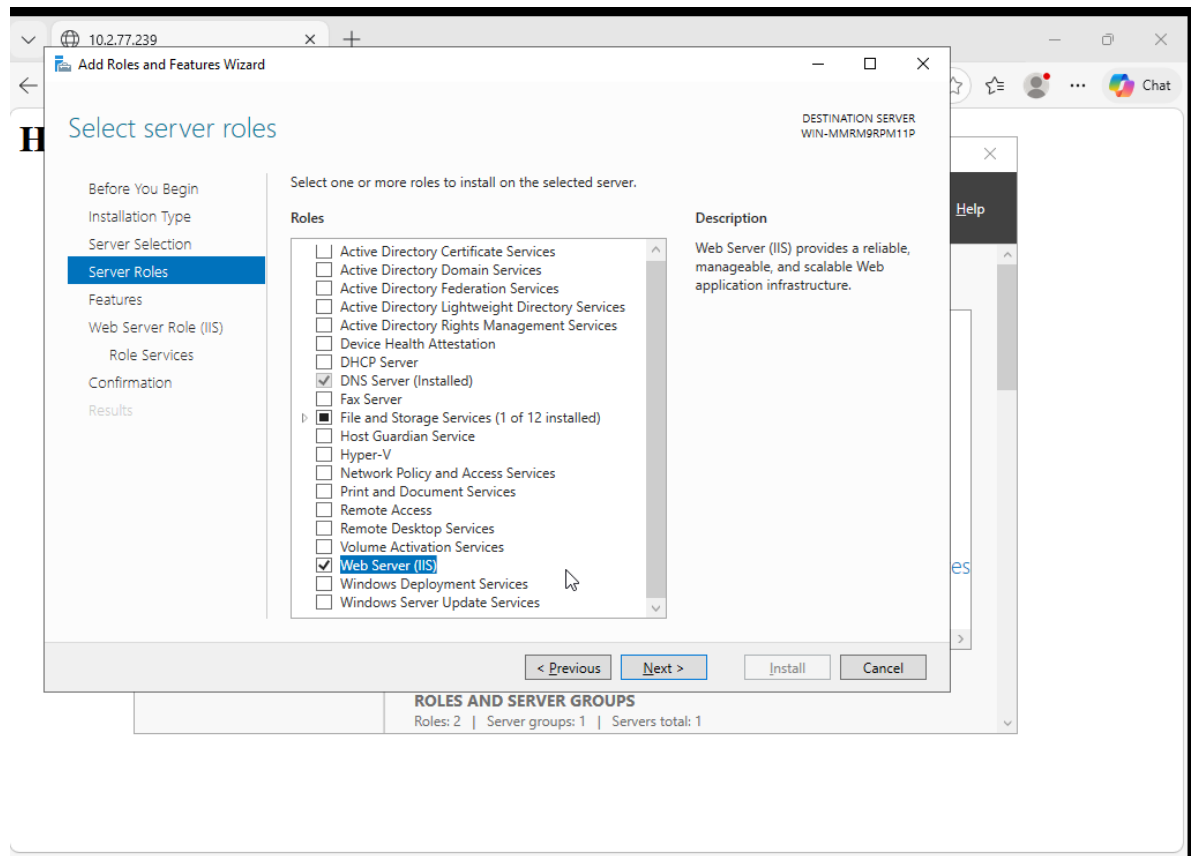
Hacemos Clic en Next hasta llegar a Server Roles



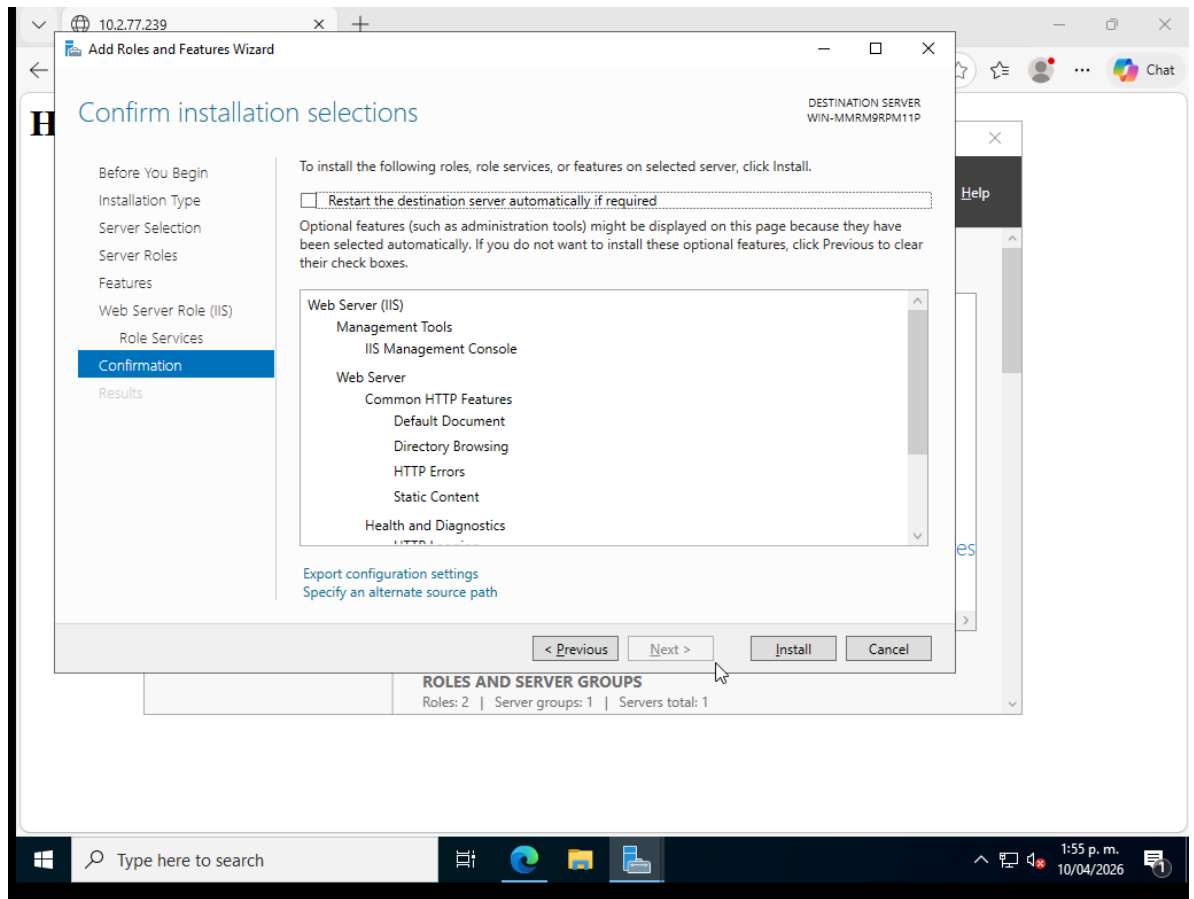
Marcamos Web Server (IIS)

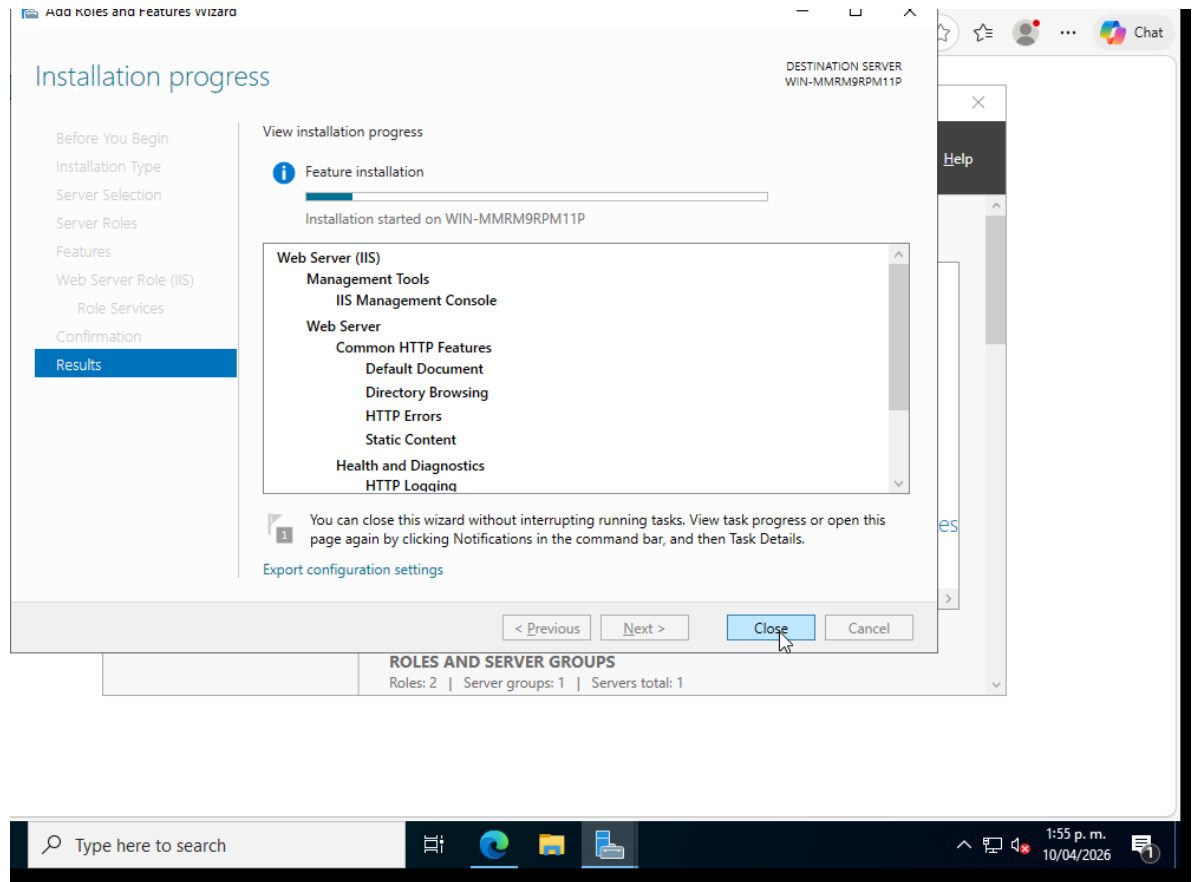


Clic en Add Features cuando pregunte

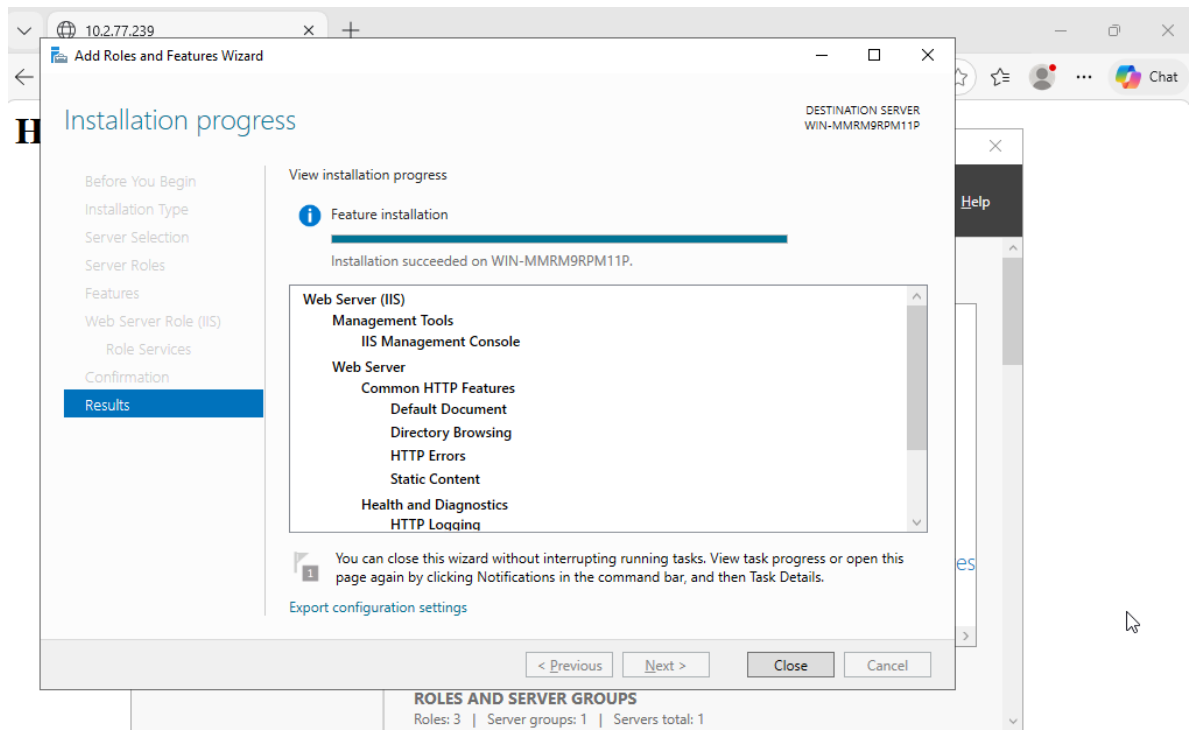


Clic en Next hasta Install





Esperamos que termine y clic en Close



Ahora crea la página web. Abre el Bloc de notas como administrador y pegamos esto:

```
<h1>Hola desde Windows Server - Student 01</h1>
```

Y guardamos en C:\inetpub\wwwroot\



Local Disk (C:)\inetpub\wwwroot

Probamos la página desde free

```
curl http://10.2.77.240
```

```
root@JK:~ # curl http://10.2.77.240
<h1>hola desde windows server- Student 01</h1>root@JK:~ # |
```

agregamos la zona en el archivo de configuración:

```
ee /usr/local/etc/namedb/named.conf
```

Y agregamos

```
zone "apache.com.co" IN {  
    type master;  
    file "/usr/local/etc/namedb/master/apache.com.co.zone";  
};
```

```
options {  
    directory "/usr/local/etc/namedb/working";  
    pid-file "/var/run/named/pid";  
  
    listen-on { any; };  
    listen-on-v6 { any; };  
  
    allow-query { any; };  
    recursion yes;  
    allow-recursion { any; };  
};  
  
// Zona directa (dominio -> IP)  
zone "angel.com.it" IN {  
    type master;  
    file "/usr/local/etc/namedb/master/angel.com.it.zone";  
    allow-transfer { 10.2.77.239; 10.2.77.240; };  
    notify yes;  
};  
  
// Zona inversa (IP -> dominio)  
zone "77.2.10.in-addr.arpa" IN {  
    type master;  
    file "/usr/local/etc/namedb/master/10.2.77.rev";  
    allow-transfer { 10.2.77.239; 10.2.77.240; };  
    notify yes;  
};  
  
zone "apache.com.co" IN {  
    type master;  
    file "/usr/local/etc/namedb/master/apache.com.co.zone";  
};  
  
press Escape (^[]) for menu
```

Ahora crea el archivo de zona:

ee /usr/local/etc/namedb/master/apache.com.co.zone

Y escribe esto adentro:

\$TTL 86400

@ IN SOA ns1.apache.com.co. admin.apache.com.co. (
2026041001 ; Serial
3600 ; Refresh
900 ; Retry
604800 ; Expire
86400) ; Minimum

@ IN NS ns1.apache.com.co.

ns1 IN A 10.2.77.241

www IN A 10.2.77.241

nginx IN A 10.2.77.239

iis IN A 10.2.77.240

```
@    IN    SOA ns1.apache.com.co. admin.apache.com.co. (
      2026041001 ; Serial
      3600      ; Refresh
      900       ; Retry
      604800    ; Expire
      86400 )   ; Minimum
```

```
@      IN    NS      ns1.apache.com.co.
ns1     IN    A       10.2.77.241
www     IN    A       10.2.77.241
nginx   IN    A       10.2.77.239
iis     IN    A       10.2.77.240
```

```
"/usr/local/etc/namedb/master/apache.com.co.zone" 13 lines, 385 characters
root@JK:~ #
```

Ahora reinicia el servicio DNS:

service named restart

```
root@JK:~ # service named restart
Stopping named.
Waiting for PIDS: 638.
Starting named.
root@JK:~ # |
```

Probamos que el dns resuelva:

nslookup www.apache.com.co 10.2.77.241

```
root@JK:~ # service named restart
Stopping named.
Waiting for PIDS: 638.
Starting named.
root@JK:~ # nslookup www.apache.com.co 10.2.77.241
Server:      10.2.77.241
Address:     10.2.77.241#53

Name:   www.apache.com.co
Address: 10.2.77.241

root@JK:~ #
```

nslookup nginx.apache.com.co 10.2.77.241

```
root@JK:~ # nslookup nginx.apache.com.co 10.2.77.241
Server:      10.2.77.241
Address:     10.2.77.241#53

Name:   nginx.apache.com.co
Address: 10.2.77.239

root@JK:~ # |
```

nslookup iis.apache.com.co 10.2.77.241

```
root@JK:~ # nslookup iis.apache.com.co 10.2.77.241
Server:      10.2.77.241
Address:     10.2.77.241#53

Name:   iis.apache.com.co
Address: 10.2.77.240

root@JK:~ #
```

Ahora para que las máquinas usen este DNS, hay que apuntarlas al servidor DNS de FreeBSD (10.2.77.241).

nano /etc/resolv.conf

```
GNU nano 6.0 /etc/resolv.conf Modified
search local1
nameserver 10.2.77.241
|

^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify
```

Guardamos y probamos

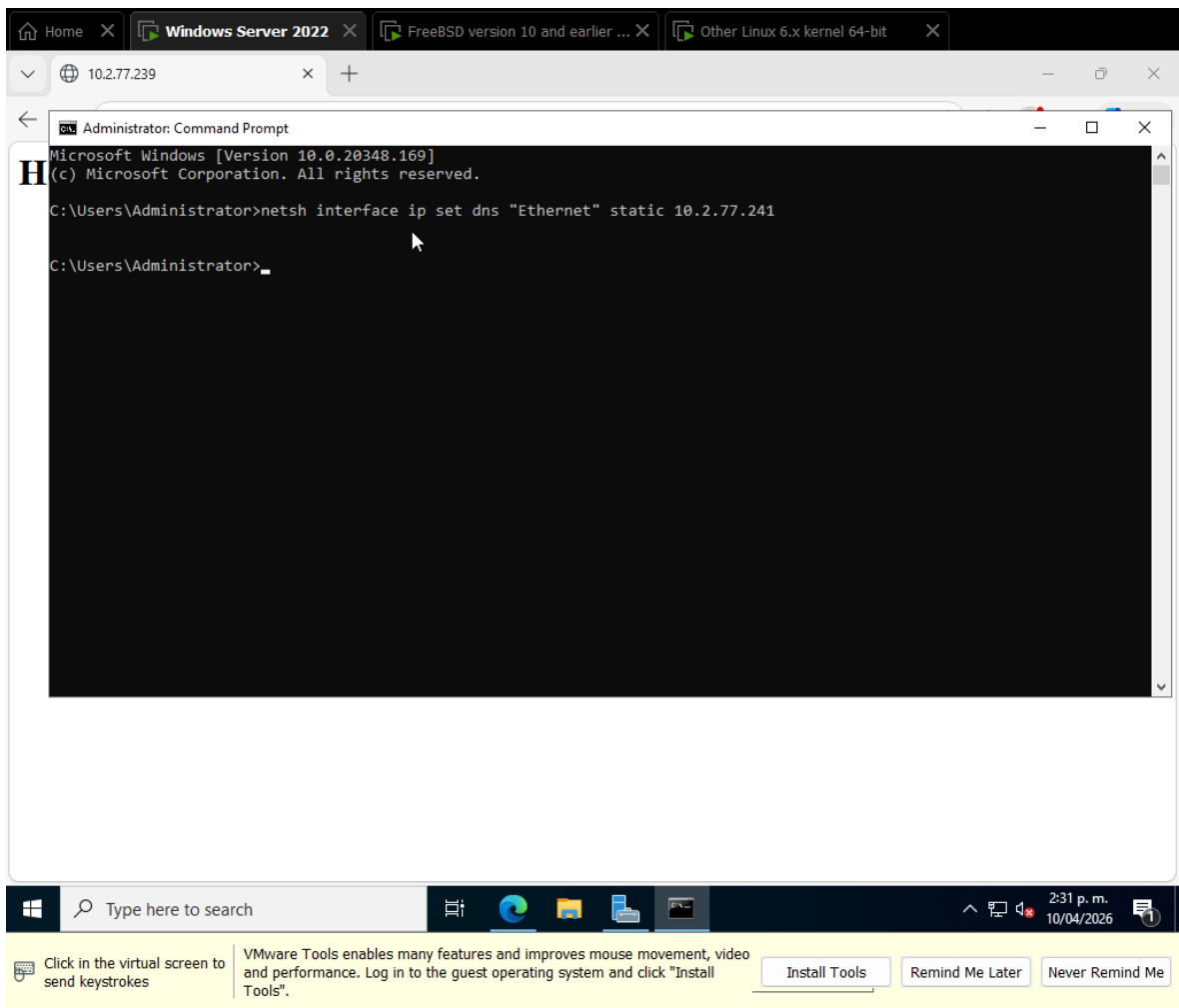
```
root@jk:~# nano /etc/resolv.conf
root@jk:~# nslookup www.apache.com.co
Server:          10.2.77.241
Address:         10.2.77.241#53

Name:   www.apache.com.co
Address: 10.2.77.241

root@jk:~# |
```

Ahora configura el DNS en Windows Server. CMD como administrador :

netsh interface ip set dns "Ethernet" static 10.2.77.241



Probamos

```
Administrator: Command Prompt
Microsoft Windows [Version 10.0.20348.169]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Administrator>netsh interface ip set dns "Ethernet" static 10.2.77.241

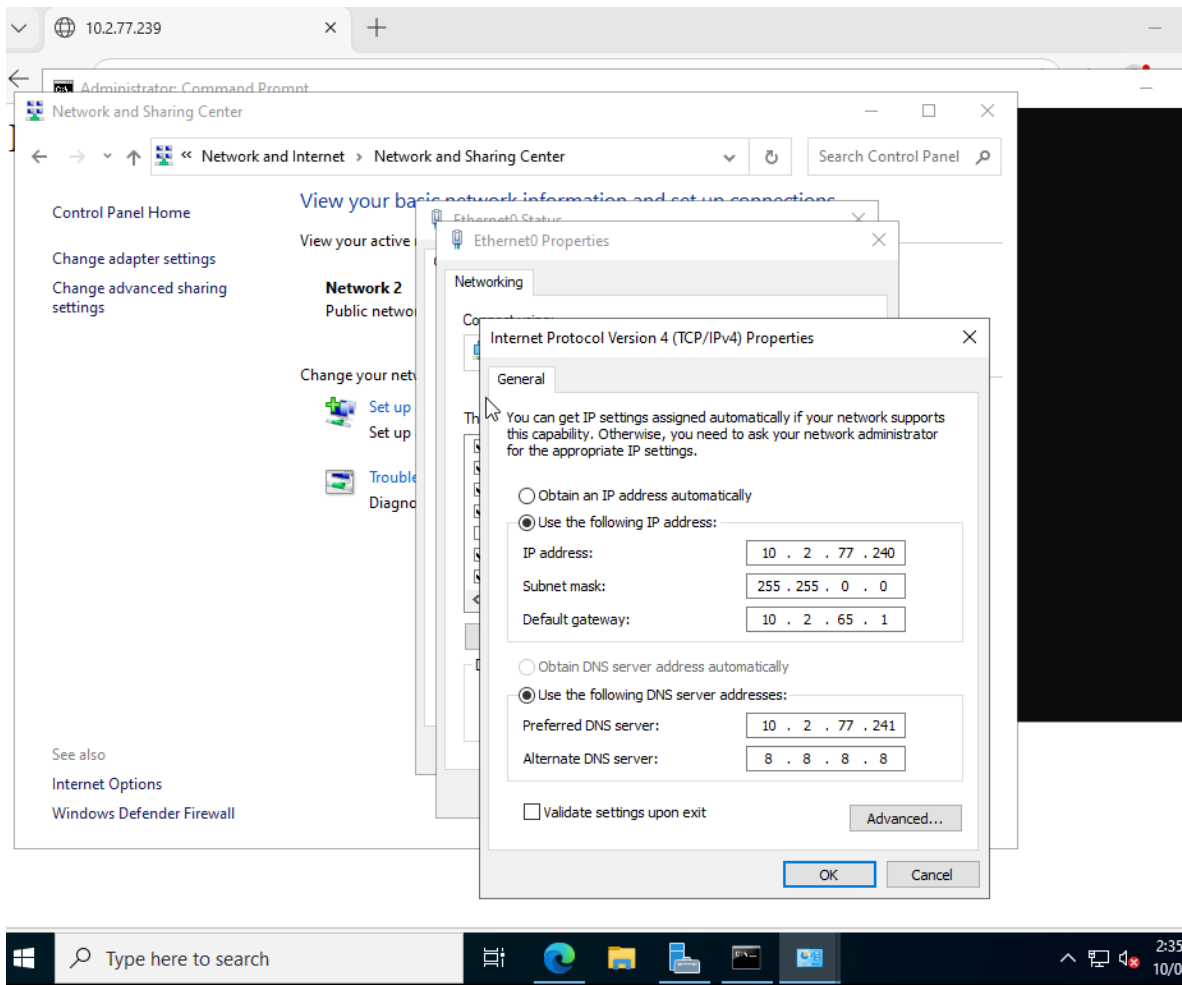
C:\Users\Administrator>nslookup www.apache.com.co
Server:  zafiro.is.escuelaing.edu.co
Address:  10.2.65.1

Non-authoritative answer:
Name:     apache.com.co
Addresses: 64:ff9b::321:82be
           64:ff9b::fc5:9421
           3.33.130.190
           15.197.148.33
Aliases:  www.apache.com.co

C:\Users\Administrator>
```



El problema es que Windows está consultando el DNS de la escuela (zafiro.is.escuelaing.edu.co) en lugar del nuestro. Debido a que lo teníamos en la creación de samba de el lab anterior



Probamos

```
C:\Users\Administrator>curl http://www.apache.com.co
<h1>Hola desde FreeBSD - Student 01</h1>

C:\Users\Administrator>
```

```
C:\Users\Administrator>curl http://nginx.apache.com.co
<h1>Hola desde Slackware - Student 01</h1>

C:\Users\Administrator>curl http://iis.apache.com.co
<h1>hola desde windows server- Student 01</h1>
C:\Users\Administrator>
```

2. Hosting Service Configuration

[For groups of 2 and 3 students]

Esta parte pide configurar 3 páginas web diferentes en el mismo servidor FreeBSD

Entonces las URLs serán:

network.kevin.com.co

security.jhonatan.org.jp

```
root@JK:~ # mkdir -p /usr/local/www/apache24/data/network.kevin.com.co
root@JK:~ # mkdir -p /usr/local/www/apache24/data/security.jhonatan.org.jp
root@JK:~ #
```

Ahora crea una página para cada sitio:

```
tee /usr/local/www/apache24/data/network.kevin.com.co/index.html << EOF
<h1>Bienvenido a network.kevin.com.co</h1>
EOF
```

```
tee /usr/local/www/apache24/data/security.jhonatan.org.jp/index.html << EOF
<h1>Bienvenido a security.jhonatan.org.jp</h1>
EOF
```

```
root@JK:~ # tee /usr/local/www/apache24/data/network.kevin.com.co/index.html
<< EOF
? <h1>Bienvenido a network.kevin.com.co</h1>
? EOF
<h1>Bienvenido a network.kevin.com.co</h1>
root@JK:~ # tee /usr/local/www/apache24/data/security.jhonatan.org.jp/index.
html << EOF
? <h1>Bienvenido a security.jhonatan.org.jp</h1>
? EOF
<h1>Bienvenido a security.jhonatan.org.jp</h1>
root@JK:~ #
```

Ahora configura los virtual hosts en Apache:

```
ee /usr/local/etc/apache24/httpd.conf
```

```
#
<IfModule ssl_module>
SSLRandomSeed startup builtin
SSLRandomSeed connect builtin
</IfModule>

Include etc/apache24/Includes/*.conf

<VirtualHost *:80>
    ServerName network.kevin.com.co
    DocumentRoot /usr/local/www/apache24/data/network.kevin.com.co
</VirtualHost>

<VirtualHost *:80>
    ServerName security.jhonatan.org.jp
    DocumentRoot /usr/local/www/apache24/data/security.jhonatan.org.jp
</VirtualHost>
press Escape (Alt) for menu
```

Reiniciamos apache

```
root@JK:~ # service apache24 restart
Performing sanity check on apache24 configuration:
AH00557: httpd: apr_sockaddr_info_get() failed for JK
AH00558: httpd: Could not reliably determine the server's fully qualified do
main name, using 127.0.0.1. Set the 'ServerName' directive globally to suppr
ess this message
Syntax OK
Stopping apache24.
Waiting for PIDS: 1051.
Performing sanity check on apache24 configuration:
AH00557: httpd: apr_sockaddr_info_get() failed for JK
AH00558: httpd: Could not reliably determine the server's fully qualified do
main name, using 127.0.0.1. Set the 'ServerName' directive globally to suppr
ess this message
Syntax OK
Starting apache24.
AH00557: httpd: apr_sockaddr_info_get() failed for JK
AH00558: httpd: Could not reliably determine the server's fully qualified do
main name, using 127.0.0.1. Set the 'ServerName' directive globally to suppr
ess this message
root@JK:~ # |
```

Ahora hay que agregar los registros DNS para los dos dominios. En FreeBSD edita el archivo de zona:

```
$TTL 86400
@ IN SOA ns1.apache.com.co. admin.apache.com.co. (
    2026041001 ; Serial
    3600       ; Refresh
    900        ; Retry
    604800     ; Expire
    86400 )    ; Minimum

@ IN NS      ns1.apache.com.co.
ns1 IN A      10.2.77.241
www IN A      10.2.77.241
nginx IN A    10.2.77.239
iis  IN A      10.2.77.240

network.kevin IN A 10.2.77.241
```

press Escape (^[]) for menu

Ahora creamos una zona nueva para jhonatan.org.jp:

ee /usr/local/etc/namedb/named.conf

```

pid-file "/var/run/named/pid";

listen-on { any; };
listen-on-v6 { any; };

allow-query { any; };
recursion yes;
allow-recursion { any; };
};

// Zona directa (dominio -> IP)
zone "angel.com.it" IN {
    type master;
    file "/usr/local/etc/namedb/master/angel.com.it.zone";
    allow-transfer { 10.2.77.239; 10.2.77.240; };
    notify yes;
};

// Zona inversa (IP -> dominio)
zone "77.2.10.in-addr.arpa" IN {
    type master;
    file "/usr/local/etc/namedb/master/10.2.77.rev";
    allow-transfer { 10.2.77.239; 10.2.77.240; };
    notify yes;
};

zone "apache.com.co" IN {

    type master;

    file "/usr/local/etc/namedb/master/apache.com.co.zone";

};

zone "jhonatan.org.jp" IN {
    type master;
    file "/usr/local/etc/namedb/master/jhonatan.org.jp.zone";
};
"/usr/local/etc/namedb/named.conf" 40 lines, 856 characters
root@JK:~ #

```

Ahora creamos el archivo de zona para jhonatan.org.jp:

ee /usr/local/etc/namedb/master/jhonatan.org.jp.zone

```
$TTL 86400
@      IN      SOA ns1.jhonatan.org.jp. admin.jhonatan.org.jp. (
        2026041001 ; Serial
        3600       ; Refresh
        900        ; Retry
        604800     ; Expire
        86400      ; Minimum

@      IN      NS      ns1.jhonatan.org.jp.
ns1     IN      A       10.2.77.241
security IN      A       10.2.77.241

"/usr/local/etc/namedb/master/jhonatan.org.jp.zone" 11 lines, 316 characters
```

Luego reiniciamos el DNS:

```
root@JK:~ # service named restart
Stopping named.
Starting named.
root@JK:~ #
```

Ahora probamos desde Windows con curl:

```
C:\Users\Administrator>curl http://network.kevin.com.co  
<h1>Bienvenido a network.kevin.com.co</h1>
```

```
C:\Users\Administrator>curl http://security.jhonatan.org.jp  
<h1>Bienvenido a security.jhonatan.org.jp</h1>
```

```
C:\Users\Administrator>
```